Citation Paraphrase Verification Log

Thesis: *Abstraction Mechanisms for Virtual Avatars*
Nidanur Günay • University of Konstanz • 2026

Every citation in the thesis has an entry here, in reading order, Ch. 1 → Ch. 7.
Each entry shows the **exact sentence from the thesis** and the **passage from the original source**.

Red box = wording quoted from the original source. **Orange** = not yet checked against the original.

Generated from the thesis, so nothing here quotes text that has been deleted.

MANDATORY WORKFLOW RULE — Citation Entry Protocol

This document is generated from the thesis, so the green boxes always match the current text. Regenerate it whenever a citation is added, changed, or removed. Each entry gives:

- **Where** it appears — chapter and section.
- **The thesis sentence** making the claim, quoted exactly.
- **The source passage** quoted from the original, or an orange note saying it has not been checked.
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Tick **Validated** once you have read the original and agree it supports the sentence.

Do not skip this step. A citation added without a corresponding entry here is unverified and must be treated as a potential error until logged.

Chapter: Introduction

#1 weidner2023 — Weidner et al. (2023) *Motivation*

Sentence in thesis: "As virtual assistants, game characters, and avatars in immersive environments increasingly involve users in sustained interactions with virtual humans, users are no longer limited to passive viewing."

Usage: First use of this source.

Original source: ""Our findings show that full-body visualizations usually outperform partial body configurations. Also, in most studies, a realistic rendering style yields better results than other rendering styles... we found that the task domain influences avatar selection as for example more task-focused activities (e.g., gaming or education) do not benefit much from highly realistic avatars or are in some cases even hampered by them." (§6 Conclusion, direct quote.)"

How to check: Read directly from `Articles/_2022_10_07__IEEE_VR__A_Survey_on_Avatar_and_Agents.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#2 mcdonnell2012 — McDonnell et al. (2012) *Motivation*

Sentence in thesis: "One of the most difficult is how realistically or abstractly such a character should be rendered."

Usage: First use of this source.

Original source: "§4: "We created 11 rendering styles in total for use in our experiments." §5.1.3 Realism: "Posthoc analysis showed that six distinct levels of realism were found" (this is the source of the previous "six" error — six is the number of statistically grouped realism levels, not the number of styles tested). Appeal: "four of the most abstract and the two most realistic styles were rated as the most appealing... and significantly more appealing than all other styles... characters that were rated halfway between abstract and realistic... were considered the least appealing." Re-assuring: "the most realistic and most abstract styles were considered the most re-assuring... HumanIll was rated as the most eerie." Trustworthy: "the most realistic Humans, and all other Toons were rated as equally trustworthy, and more trustworthy than all other styles." §7 Discussion: "These results are very much in line with the Uncanny Valley theory. Contrary to the theory, however, was the fact that our most realistic character was often rated as appealing or pleasant as our cartoon characters. The drop in appeal for our stimuli came from characters that were rated as unfamiliar or those that were in the middle of our 'abstract to realistic' scale.""

How to check: Read directly from `Articles/McDonnell, R., Breidt, M., and Bulthoff, H. H. (2012). Render me Real? Investigating.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#3 natarajan2020 — Natarajan & Gombolay (2020) *Motivation*

Sentence in thesis: "Furthermore, an avatar's design and behaviour have been shown to increase users' confidence in its guidance and their willingness to follow it."

Usage: First use of this source.

Original source: "Abstract: "Results from the 4x4x2x2 study show that behavior and anthropomorphism of the agent are the most significant factors in predicting the trust and compliance with the robot." §3.4 H1: "The perceived anthropomorphism of an agent will have a positive correlation with the user's trust... we hypothesize that increased perception of anthropomorphism of an agent by users will lead to an increase in compliance with the agent's recommendations and overall trust in the agent." §4.1 Results: "Results from repeated measures ANOVA (rANOVA) show high statistical significance for perceived anthropomorphism ($p<0.001$, $F(1, 720)=17.6276$). Further, we obtained a positive coefficient (0.29335) for anthropomorphism... indicating that trust and anthropomorphism are positively correlated." §4.2 H2 (behavior): "behavior of the agent plays a significant role in affecting user's trust" ($p<0.001$, $F(4,670)=73.5384$)."

How to check: Read directly from `Articles/Natarajan-Effects of Anthropomorphism and Accountability on Trust inHuman Robot Interaction.pdf`. Confirm this passage supports the sentence above.

✔ **Validated** — I have read the original and agree it supports the sentence.

#4 mori1970 — Mori (1970) *Motivation*

Sentence in thesis: "Mori termed this effect the uncanny valley: comfort rises with human likeness only up to a point, then drops sharply for entities that appear almost but not quite human, before recovering near perfect likeness."

Usage: First use of this source.

Original source: "p. 1 ("Valley of familiarity"): "I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. I call this relation the 'uncanny valley.'" Figure 1 (the familiarity-vs-human-likeness graph, x-axis 0–100% human likeness): the curve rises from industrial robot through humanoid robot/stuffed animal, drops sharply through the trough (labelled "corpse" and "prosthetic hand," positioned close to the 100% end of the axis), then rises again to "healthy person" at 100% human likeness, the highest point on the graph (the "second peak"). p. 2 ("The significance of the uncanny"): "In Figure 1, a healthy person is at the top of the second peak.""

How to check: Read directly from `Articles/Uncaney/uncanny1970.pdf`. Confirm this passage supports the sentence above.

✔ **Validated** — I have read the original and agree it supports the sentence.

#5 mcdonnell2012 — McDonnell et al. (2012) *Motivation*

Sentence in the-sis: "This holds for CG characters generally."

Usage: Also cited at entry #2.

Original source: "§4: "We created 11 rendering styles in total for use in our experiments." §5.1.3 Realism: "Posthoc analysis showed that six distinct levels of realism were found" (this is the source of the previous "six" error — six is the number of statistically grouped realism levels, not the number of styles tested). Appeal: "four of the most abstract and the two most realistic styles were rated as the most appealing... and significantly more appealing than all other styles... characters that were rated halfway between abstract and realistic... were considered the least appealing." Re-assuring: "the most realistic and most abstract styles were considered the most re-assuring... HumanIll was rated as the most eerie." Trustworthy: "the most realistic Humans, and all other Toons were rated as equally trustworthy, and more trustworthy than all other styles." §7 Discussion: "These results are very much in line with the Uncanny Valley theory. Contrary to the theory, however, was the fact that our most realistic character was often rated as appealing or pleasant as our cartoon characters. The drop in appeal for our stimuli came from characters that were rated as unfamiliar or those that were in the middle of our 'abstract to realistic' scale.""

How to check: Same source as entry #2.

Validated — I have read the original and agree it supports the sentence.

#6 canales2024 — Canales et al. (2024) *Motivation*

Sentence in the-sis: "Avatars fall within this category, and their stylisation can affect users' behavioural choice."

Usage: First use of this source.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Read directly from `Articles/trust/Avatar_Stylization_and_Trust_IEEE_VR_-2024.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#7 rahman2024 — Rahman et al. (2024) *Motivation*

Sentence in thesis: "This question is becoming increasingly relevant as virtual avatars move beyond research settings into widely used consumer applications such as gaming and advertising."

Usage: First use of this source.

Original source: "p. 2 of 46: "Companies like Nike have introduced Nikeland in the Metaverse through platforms like Roblox. Similarly, IKEA has introduced a customer-focused avatar named Anna, who can address inquiries regarding IKEA's products, services, and operating hours." Same page: "...the growing importance of avatars in online services (e.g., banking, gaming, education, shopping, m-health, etc.)... the usage of avatars is not limited to a particular sector, like banking or hotel; it is seen as a huge prospect in the virtual gaming, smartphone, and fashion industries." Same page (advertising half): "...the famous fashion brand Calvin Klein has deployed Lil Miquela, a virtual influencer (another form of avatar); interestingly, Lil Miquela has more than 2.4 million followers on Instagram.""

How to check: Read directly from `Articles/Navigating_Avatar-Based_Marketing_From_Value_-Aspec.pdf`. Confirm this passage supports the sentence above.

✔ **Validated** — I have read the original and agree it supports the sentence.

#8 canales2024 — Canales et al. (2024) *Problem Statement*

Sentence in thesis: "A recent avatar stylisation study focused on perceptual outcomes rather than the practical challenges of production integration."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#9 isenberg2003 — Isenberg et al. (2003) *Problem Statement*

Sentence in the-sis: "Classical NPR techniques share this same gap, since they are typically demonstrated in rendering environments where developers have direct control over rendering operations and asset creation."

Usage: First use of this source.

Original source: "[Classification section, p. 29:] "we distinguish between image space algorithms that only operate on image buffers, hybrid algorithms that perform manipulations in object space but yield the silhouette in an image buffer, and object space algorithms that perform all calculations in object space." [Image space algorithms section, p. 29:] "Image space algorithms exploit discontinuities in the image buffer(s) that result from conventional rendering and extract them using image-processing methods.""

How to check: Read directly from `Articles/NPR/Isenberg_2003_ADG.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#10 zell2015 — Zell et al. (2015) *Problem Statement*

Sentence in the-sis: "This is because stronger abstraction typically reduces the perceived intensity of facial expressions, even where basic recognition is preserved."

Usage: First use of this source.

Original source: "Main Findings list, p. 184:2: "The perceived intensity of expressions decreases with realism of shape, but is nearly independent of material stylization.""

How to check: Read directly from `Articles/NPR/To Stylize or not to Stylize?_Zell.pdf`. Confirm this passage supports the sentence above.

✔ **Validated** — I have read the original and agree it supports the sentence.

#11 wallraven2008 — Wallraven et al. (2008) *Problem Statement*

Sentence in the-sis: "This is because stronger abstraction typically reduces the perceived intensity of facial expressions, even where basic recognition is preserved."

Usage: First use of this source.

Original source: "§4 Summary, p. 23:19: "In addition, in Experiment 2, we found clear effects of shape blurring on both intensity and sincerity judgments. This demonstrates that, even though reduced shape information does not affect recognition, it does influence the perceived quality of the expression.""

How to check: Read directly from `Articles/NPR/Evaluating the perceptual realism of animated facial expressions. .pdf`. Confirm this passage supports the sentence above.

✔ **Validated** — I have read the original and agree it supports the sentence.

Chapter: Background

#12 gonzalez2018 — Gonzalez & Woods (2018) *Notation and Conventions*

Sentence in the-sis: "It transposes the row and column ordering adopted in the image processing literature."

Usage: First use of this source.

Original source: "[p. 18, §1.1 "What Is Digital Image Processing?":] "Note that a digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels, and pixels." [p. 66, Ch. 2, Eq. (2-9)–(2-10):] the image is formally represented as an $M \times N$ array $f(x, y)$, and "[e]ach element of this array is called an image element, picture element, pixel, or pel." [p. 702, §10.2 "Point, Line, and Edge Detection":] "Edge pixels are pixels at which the intensity of an image changes abruptly, and edges (or edge segments) are sets of connected edge pixels.""

How to check: Read directly from `Articles/Gonzales,Woods-Digital.Image.Processing.4th.Edition.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#13 akenine2018 — Akenine-Möller et al. (2018) *Rendering Concepts*

Sentence in the-sis: "Its result is written into a render target, an off-screen image buffer."

Usage: First use of this source.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: Akenine-Möller et al., *Real-Time Rendering*, 4th ed., A K Peters/CRC Press 2018, ISBN 978-1-138-62700-0"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#14 unityurp — Unity Technologies (2026) *Rendering Concepts*

Sentence in the-sis: "Unity's Universal Render Pipeline (URP) schedules such injected work through a renderer feature."

Usage: First use of this source.

Original source: ""Use the `ScriptableRenderFeature` API to insert a Scriptable Render Pass into the Universal Render Pipeline (URP) frame rendering loop." The page further specifies overriding `Create` to instantiate the pass and `AddRenderPasses` to enqueue it: "Use the `EnqueuePass` API to inject the Scriptable Render Pass into the frame rendering loop.""

How to check: Read directly from Unity 6 URP manual page — fetched and read. Confirm this passage supports the sentence above.

✔ **Validated** — I have read the original and agree it supports the sentence.

#15 akenine2018 — Akenine-Möller et al. (2018) *Rendering Concepts*

Sentence in the-sis: "It is held in separate texture maps under physically based rendering (PBR)."

Usage: Also cited at entry #13.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: Akenine-Möller et al., *Real-Time Rendering*, 4th ed., A K Peters/CRC Press 2018, ISBN 978-1-138-62700-0"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#16 luebke2003 — Luebke et al. (2003) *Rendering Concepts*

Sentence in the-sis: "A level of detail (LOD) system substitutes a simplified mesh as the character moves further away."

Usage: First use of this source.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: D. Luebke et al., *Level of Detail for 3D Graphics*, Morgan Kaufmann 2003, ISBN 978-1-55860-838-2"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#17 magnenat1988 — Magnenat-Thalmann et al. (1988) *Rendering Concepts*

Sentence in the-sis: "A skeleton deforms the body, and every vertex carries bone weights that blend the transforms acting on it."

Usage: First use of this source.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: N. Magnenat-Thalmann, R. Laperrière, D. Thalmann, "Joint-Dependent Local Deformations for Hand Animation and Object Grasping," *Graphics Interface '88*, pp. 26–33"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#18 lewis2014 — Lewis et al. (2014) *Rendering Concepts*

Sentence in the-sis: "They hold alternative vertex positions that are interpolated against the base mesh."

Usage: First use of this source.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: J. P. Lewis et al., "Practice and Theory of Blendshape Facial Models," *Eurographics 2014 STARs*, pp. 199–218, DOI 10.2312/egst.20141042 (free at graphics.cs.uh.edu)"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#19 mayer1995 — Mayer et al. (1995) *Avatars, Trust, and Eeriness*

Sentence in the-sis: "Perceived trustworthiness decomposes into ability, benevolence, and integrity."

Usage: First use of this source.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/Mayer-IntegrativeModelOrganizational-1995.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#20 mcknight2001 — McKnight & Chervany (2001) *Avatars, Trust, and Eeriness*

Sentence in thesis: "A fourth dimension, predictability, joins the three once they are operationalised as trusting beliefs for human-computer interaction."

Usage: First use of this source.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/trust/What Trust Means in E-Commerce Customer Relationships.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#21 bartneck2009 — Bartneck et al. (2009) *Avatars, Trust, and Eeriness*

Sentence in thesis: "The Godspeed questionnaire supplies five indices totalling 24 semantic differential items, each rated on a five point scale between two bipolar anchors."

Usage: First use of this source.

Original source: "[Abstract:] "The results have been distilled into five consistent questionnaires using semantic differential scales." [§2:] "In semantic differential scales the respondent is asked to indicate his or her position on a scale between two bipolar words, the anchors. . . a semantic differential format may effectively reduce acquiescence bias without lowering psychometric quality." Figure 1 shows a five point scale ("Strong 1 2 3 4 5 Weak"). Item counts per index are as reported by Ho & MacDorman (2010, §2.1.2), who used the instrument directly: "composed of five indices and 24 semantic differential items. The anthropomorphism index has five items, the animacy index has six items, the likeability index has five items, the perceived intelligence index has five items, and the perceived safety index has three items.""

How to check: Read directly from `Articles/trust/Bartneck_2009_Measurement Instruments for the Anthropomorphism.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#22 ho2010 — Ho & MacDorman (2010) *Avatars, Trust, and Eeriness*

Sentence in the-sis: "A dedicated instrument fills that gap with eight semantic differential items for eeriness alongside six for perceived humanness."

Usage: First use of this source.

Original source: "[Abstract:] "Phase 2 of this study developed alternative indices with non-significant correlations ($p > .05$) between the proposed y-axis eeriness and x-axis perceived humanness ($r = .02$)." [§3.3.2 Eeriness index:] "Therefore, the final version of the eeriness index would retain eight items that explained 62.04% of the variance and held an overall internal reliability of .74." [§3.3.1 Humanness index:] "Therefore, the final version of the humanness index would retain six items.""

How to check: Read directly from `Articles/trust/Ho2010UncannyValleyIndices.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#23 gonzalez2018 — Gonzalez & Woods (2018) *Edge Detection*

Sentence in the-sis: "An edge is conventionally defined as a location at which this intensity changes rapidly."

Usage: Also cited at entry #12.

Original source: "[p. 18, §1.1 "What Is Digital Image Processing?":] "Note that a digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels, and pixels." [p. 66, Ch. 2, Eq. (2-9)–(2-10):] the image is formally represented as an $M \times N$ array $f(x, y)$, and "[e]ach element of this array is called an image element, picture element, pixel, or pel." [p. 702, §10.2 "Point, Line, and Edge Detection":] "Edge pixels are pixels at which the intensity of an image changes abruptly, and edges (or edge segments) are sets of connected edge pixels.""

How to check: Same source as entry #12.

Validated — I have read the original and agree it supports the sentence.

#24 gonzalez2018 — Gonzalez & Woods (2018) *Edge Detection → Gradient-Based Detection*

Sentence in thesis: "It estimates the two directional derivatives with a pair of 3×3 kernels."

Usage: Also cited at entry #12.

Original source: "[p. 18, §1.1 "What Is Digital Image Processing?":] "Note that a digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels, and pixels." [p. 66, Ch. 2, Eq. (2-9)–(2-10):] the image is formally represented as an $M \times N$ array $f(x, y)$, and "[e]ach element of this array is called an image element, picture element, pixel, or pel." [p. 702, §10.2 "Point, Line, and Edge Detection":] "Edge pixels are pixels at which the intensity of an image changes abruptly, and edges (or edge segments) are sets of connected edge pixels.""

How to check: Same source as entry #12.

Validated — I have read the original and agree it supports the sentence.

#25 gonzalez2018 — Gonzalez & Woods (2018) *Edge Detection → Gradient-Based Detection*

Sentence in thesis: "The central row and column of each Sobel kernel carry weights of ± 2, whereas the simpler Prewitt operator weights all three rows and columns equally at ± 1."

Usage: Also cited at entry #12.

Original source: "[p. 18, §1.1 "What Is Digital Image Processing?":] "Note that a digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels, and pixels." [p. 66, Ch. 2, Eq. (2-9)–(2-10):] the image is formally represented as an $M \times N$ array $f(x, y)$, and "[e]ach element of this array is called an image element, picture element, pixel, or pel." [p. 702, §10.2 "Point, Line, and Edge Detection":] "Edge pixels are pixels at which the intensity of an image changes abruptly, and edges (or edge segments) are sets of connected edge pixels.""

How to check: Same source as entry #12.

Validated — I have read the original and agree it supports the sentence.

#26 roberts1963 — Roberts (1963) *Edge Detection → Gradient-Based Detection*

Sentence in the-sis: "A smaller alternative is the Roberts Cross operator, which differentiates along the two diagonals of a 2×2 neighbourhood."

Usage: First use of this source.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: L. G. Roberts, *Machine Perception of Three-Dimensional Solids*, PhD thesis, MIT 1963 (Lincoln Lab TR-315), via MIT DSpace"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#27 gonzalez2018 — Gonzalez & Woods (2018) *Edge Detection → Gradient-Based Detection*

Sentence in the-sis: "Its edge magnitude is commonly approximated as $|R_1 * I| + |R_2 * I|$, which avoids the square root of *[cross-ref]*."

Usage: Also cited at entry #12.

Original source: "[p. 18, §1.1 "What Is Digital Image Processing?":] "Note that a digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels, and pixels." [p. 66, Ch. 2, Eq. (2-9)–(2-10):] the image is formally represented as an $M \times N$ array $f(x, y)$, and "[e]ach element of this array is called an image element, picture element, pixel, or pel." [p. 702, §10.2 "Point, Line, and Edge Detection":] "Edge pixels are pixels at which the intensity of an image changes abruptly, and edges (or edge segments) are sets of connected edge pixels.""

How to check: Same source as entry #12.

Validated — I have read the original and agree it supports the sentence.

#28 itu709 — International Telecommunication Union (2015) *Edge Detection → Gradient-Based Detection*

Sentence in the-sis: "Gradient computation is accordingly performed on a single luma channel formed as a weighted sum of the red, green, and blue components: L = 0.2126,R + 0.7152,G + 0.0722,B, following the coefficients defined in ITU-R Recommendation BT.709 for high-definition video."

Usage: First use of this source.

Original source: "[Item 3.2, "Derivation of luminance signal E'_Y":] $E'_Y = 0.2126\,E'_R + 0.7152\,E'_G + 0.0722\,E'_B$."

How to check: Read directly from ITU-R BT.709-6 official PDF (itu.int) — fetched and read. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#29 marr1980 — Marr & Hildreth (1980) *Edge Detection → Gaussian Prefiltering*

Sentence in the-sis: "Marr and Hildreth established that reliable edge detection requires low-pass smoothing before the gradient is computed."

Usage: First use of this source.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: D. Marr and E. Hildreth, "Theory of Edge Detection," *Proc. R. Soc. Lond. B* 207, 1980, pp. 187–217, DOI 10.1098/rspb.1980.0020"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#30 canny1986 — Canny (1986) *Edge Detection → Gaussian Prefiltering*

Sentence in the-sis: "Canny built an influential detector on this principle."

Usage: First use of this source.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: J. Canny, "A Computational Approach to Edge Detection," *IEEE TPAMI* 8(6), 1986, pp. 679–698, DOI 10.1109/TPAMI.1986.4767851"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#31 kuwahara1976 — Kuwahara et al. (1976) *Structure-Preserving Filtering*

Sentence in the-sis: "It was originally developed to smooth angiocardiographic images without losing the boundaries between anatomical regions."

Usage: First use of this source.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: M. Kuwahara et al., "Processing of RI-Angiocardiographic Images," in *Digital Processing of Biomedical Images*, Plenum Press 1976, pp. 187–203, DOI 10.1007/978-1-4684-0769-3_13"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#32 kyprianidis2009 — Kyprianidis et al. (2009) *Structure-Preserving Filtering*

Sentence in the-sis: "Interior regions therefore flatten towards uniform fills while boundaries stay sharp, and the result resembles painterly brushwork."

Usage: First use of this source.

Original source: "[Abstract:] "We present a non-photorealistic rendering technique to transform color images and videos into painterly abstractions. It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor." [§3.1 "Orientation and Anisotropy Estimation":] "We calculate the structure tensor directly from the RGB values of the input. . . The structure tensor of f is then defined as" the matrix of pairwise dot products $f_x \cdot f_x$, $f_x \cdot f_y$, $f_y \cdot f_y$, subsequently smoothed with a Gaussian; "Local orientation and a measure for the anisotropy are then derived from the eigenvalues and eigenvectors of the smoothed structure tensor.""

How to check: Read directly from `Articles/NPR/anisotropic_kuwahara.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#33 kyprianidis2009 — Kyprianidis et al. (2009) *Structure-Preserving Filtering*

Sentence in the-sis: "Kyprianidis et al. recovered it by aligning the sampling window with local image structure, which they estimate through the structure tensor."

Usage: Also cited at entry #32.

Original source: "[Abstract:] "We present a non-photorealistic rendering technique to transform color images and videos into painterly abstractions. It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor." [§3.1 "Orientation and Anisotropy Estimation":] "We calculate the structure tensor directly from the RGB values of the input. . . The structure tensor of f is then defined as" the matrix of pairwise dot products $f_x \cdot f_x$, $f_x \cdot f_y$, $f_y \cdot f_y$, subsequently smoothed with a Gaussian; "Local orientation and a measure for the anisotropy are then derived from the eigenvalues and eigenvectors of the smoothed structure tensor.""

How to check: Same source as entry #32.

Validated — I have read the original and agree it supports the sentence.

Chapter: Related Work

#34 mori1970 — Mori (1970) *The Uncanny Valley and Avatar Realism*

Sentence in thesis: "Only then is comfort recovered."

Usage: Also cited at entry #4.

Original source: "p. 1 ("Valley of familiarity"): "I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. I call this relation the 'uncanny valley.'" Figure 1 (the familiarity-vs-human-likeness graph, x-axis 0–100% human likeness): the curve rises from industrial robot through humanoid robot/stuffed animal, drops sharply through the trough (labelled "corpse" and "prosthetic hand," positioned close to the 100% end of the axis), then rises again to "healthy person" at 100% human likeness, the highest point on the graph (the "second peak"). p. 2 ("The significance of the uncanny"): "In Figure 1, a healthy person is at the top of the second peak.""

How to check: Same source as entry #4.

Validated — I have read the original and agree it supports the sentence.

#35 mori1970 — Mori (1970) *The Uncanny Valley and Avatar Realism*

Sentence in thesis: "Each looks human in most respects, but each has one flaw that people notice instantly."

Usage: Also cited at entry #4.

Original source: "p. 1 ("Valley of familiarity"): "I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. I call this relation the 'uncanny valley.'" Figure 1 (the familiarity-vs-human-likeness graph, x-axis 0–100% human likeness): the curve rises from industrial robot through humanoid robot/stuffed animal, drops sharply through the trough (labelled "corpse" and "prosthetic hand," positioned close to the 100% end of the axis), then rises again to "healthy person" at 100% human likeness, the highest point on the graph (the "second peak"). p. 2 ("The significance of the uncanny"): "In Figure 1, a healthy person is at the top of the second peak.""

How to check: Same source as entry #4.

Validated — I have read the original and agree it supports the sentence.

#36 mori1970 — Mori (1970) *The Uncanny Valley and Avatar Realism (figure caption)*

Sentence in thesis: "The vertical axis represents comfort level (*shinwakan*). The horizontal axis represents degree of human likeness. The solid curve describes affective response to still figures, while the dashed curve describes figures in motion."

Usage: Also cited at entry #4.

Original source: "p. 1 ("Valley of familiarity"): "I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. I call this relation the 'uncanny valley.'" Figure 1 (the familiarity-vs-human-likeness graph, x-axis 0–100% human likeness): the curve rises from industrial robot through humanoid robot/stuffed animal, drops sharply through the trough (labelled "corpse" and "prosthetic hand," positioned close to the 100% end of the axis), then rises again to "healthy person" at 100% human likeness, the highest point on the graph (the "second peak"). p. 2 ("The significance of the uncanny"): "In Figure 1, a healthy person is at the top of the second peak.""

How to check: Same source as entry #4.

Validated — I have read the original and agree it supports the sentence.

#37 macdorman2006 — MacDorman & Ishiguro (2006) *The Uncanny Valley and Avatar Realism*

Sentence in thesis: "This is because androids can be positioned at any point along the human likeness continuum with experimental control."

Usage: First use of this source.

Original source: "p. 297 abstract: "The development of robots that closely resemble human beings can contribute to cognitive research. An android provides an experimental apparatus that has the potential to be controlled more precisely than any human actor." p. 298: "An android is defined to be 'an artificial system designed with the ultimate goal of being indistinguishable from humans in its external appearance and behavior' (MacDorman & Ishiguro, 2005).""

How to check: Read directly from `Articles/MacDorman2006AndroidScience.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#38 mori1970 — Mori (1970) *The Uncanny Valley and Avatar Realism*

Sentence in the-sis: "An android positioned at such an ambiguous point falls into the region Mori's curve identifies as the sharpest drop in comfort."

Usage: Also cited at entry #4.

Original source: "p. 1 ("Valley of familiarity"): "I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. I call this relation the 'uncanny valley.'" Figure 1 (the familiarity-vs-human-likeness graph, x-axis 0–100% human likeness): the curve rises from industrial robot through humanoid robot/stuffed animal, drops sharply through the trough (labelled "corpse" and "prosthetic hand," positioned close to the 100% end of the axis), then rises again to "healthy person" at 100% human likeness, the highest point on the graph (the "second peak"). p. 2 ("The significance of the uncanny"): "In Figure 1, a healthy person is at the top of the second peak.""

How to check: Same source as entry #4.

Validated — I have read the original and agree it supports the sentence.

#39 macdorman2006 — MacDorman & Ishiguro (2006) *The Uncanny Valley and Avatar Realism (figure caption)*

Sentence in the-sis: "An android (left, Replicee R1) alongside a mechanical-looking humanoid robot (right, Eveliee P1), developed at Hiroshi Ishiguro's Intelligent Robotics Laboratory."

Usage: Also cited at entry #37.

Original source: "p. 297 abstract: "The development of robots that closely resemble human beings can contribute to cognitive research. An android provides an experimental apparatus that has the potential to be controlled more precisely than any human actor." p. 298: "An android is defined to be 'an artificial system designed with the ultimate goal of being indistinguishable from humans in its external appearance and behavior' (MacDorman & Ishiguro, 2005)."

How to check: Same source as entry #37.

Validated — I have read the original and agree it supports the sentence.

#40 mcdonnell2009 — McDonnell et al. (2009) *The Uncanny Valley and Avatar Realism*

Sentence in thesis: "This independence remained consistent across photorealistic, cartoon, and stylised skeletal representations."

Usage: First use of this source.

Original source: ""The perception of emotional actions is highly robust and to the most part independent of the character's body, so long as form is present." (Physical PDF confirmed: McDonnell09_HappyZombies.pdf — ACM Reference: McDonnell et al., 2009, DOI: 10.1145/1577755.1577757.)"

How to check: Read directly from `Articles/McDonnell09_HappyZombies.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#41 mcdonnell2012 — McDonnell et al. (2012) *The Uncanny Valley and Avatar Realism*

Sentence in thesis: "The authors thereby established that stylised rendering is not merely a technical shortcut but a principled perceptual strategy for sidestepping the uncanny valley."

Usage: Also cited at entry #2.

Original source: "§4: "We created 11 rendering styles in total for use in our experiments." §5.1.3 Realism: "Posthoc analysis showed that six distinct levels of realism were found" (this is the source of the previous "six" error — six is the number of statistically grouped realism levels, not the number of styles tested). Appeal: "four of the most abstract and the two most realistic styles were rated as the most appealing... and significantly more appealing than all other styles... characters that were rated halfway between abstract and realistic... were considered the least appealing." Re-assuring: "the most realistic and most abstract styles were considered the most re-assuring... HumanIII was rated as the most eerie." Trustworthy: "the most realistic Humans, and all other Toons were rated as equally trustworthy, and more trustworthy than all other styles." §7 Discussion: "These results are very much in line with the Uncanny Valley theory. Contrary to the theory, however, was the fact that our most realistic character was often rated as appealing or pleasant as our cartoon characters. The drop in appeal for our stimuli came from characters that were rated as unfamiliar or those that were in the middle of our 'abstract to realistic' scale.""

How to check: Same source as entry #2.

Validated — I have read the original and agree it supports the sentence.

#42 dill2012 — Dill et al. (2012) *The Uncanny Valley and Avatar Realism*

Sentence in thesis: "Motion amplified this effect for every stimulus tested."

Usage: First use of this source.

Original source: ""The data produced a chart closely following the original uncanny valley curve. Characters in the near-human range showed the characteristic negative response. The video curve showed more pronounced valley effects for near-human characters than the still-image curve." (Article summary; physical PDF is the IVA 2012 proceedings volume — Evaluation of the Uncanny Valley in CG Characters.pdf.)"

How to check: Read directly from `Articles/NPR/Evaluation of the Uncanny Valley in CG Characters.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#43 seymour2021 — Seymour et al. (2021) *The Uncanny Valley and Avatar Realism*

Sentence in thesis: "None of the eighteen participants who interacted with it reported an uncanny or aversive response, and the avatar was also rated as having greater affinity, trustworthiness, and preference than a caricature counterpart."

Usage: First use of this source.

Original source: "Abstract: "Our results suggest that it is now possible to cross the uncanny valley with human-realistic avatars rendered in real time." Title: "Have We Crossed the Uncanny Valley?" Table 5 (Observation Results): "Uncanniness of Digital MIKE: 0 of 18" guests displaying an aversive response. p. 15: "All but three participants preferred the more human-realistic avatar to the caricature avatar." H1–H3 (p. 5): greater affinity, greater trustworthiness, and preference for the human-realistic avatar over the caricature, all supported per the abstract's summary of results."

How to check: Read directly from `Articles/Seymour, Yuan, Dennis & Riemer, "Have We Crossed the Uncanny Valley?" (2021).pdf`. Confirm this passage supports the sentence above.

✔ **Validated** — I have read the original and agree it supports the sentence.

#44 kuffner2024 — Kuffner dos Anjos & Madeiras Pereira (2024) *The Uncanny Valley and Avatar Realism*

Sentence in thesis: "Kuffner dos Anjos and Pereira observed that realistic mesh avatars in third-person VR perspective produced pronounced uncanny effects when animation quality was constrained by sensor resolution, while abstract representations avoided the valley entirely in the same viewing condition."

Usage: First use of this source.

Original source: ""Mesh in 3PP was the worst performing condition on the obstacle-crouching task. The authors attribute this to the uncanny valley: the realistic appearance sets higher behavioural expectations that the imperfect Kinect animation fails to meet when viewed from outside the body. Abstract in 3PP produced higher body ownership than Mesh in 3PP." (Article summary; PDF confirmed: Effects of Realism and Representation on Self-Embodied Avatars.pdf.)"

How to check: Read directly from `Articles/Effects of Realism and Representation on Self-Embodied Avatars in Immersive Virtual Environments.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#45 tao2025 — Tao et al. (2025) *The Uncanny Valley and Avatar Realism*

Sentence in thesis: "None of the eeriness comparisons reached significance, although the cumulative ranking placed medium realism as the most eerie. High realism avatars were rated significantly more trustworthy than low realism avatars, while medium realism differed significantly from neither."

Usage: First use of this source.

Original source: "**Primary source read** (`Articles/Tao-Impact.pdf`). §3.6: "Eight studies (61.5%) included comparisons of trustworthiness. The NMA results indicated that the HRA received significantly higher ratings than the LRA ($p = 0.046$). No significant differences were observed between the MRA group and either the HRA or LRA groups." §3.7: "Five studies (38.4%) included comparisons of eeriness . . . these differences were not statistically significant at the $\alpha = 0.05$ level. No significant difference was observed between the MRA and HRA either.""

How to check: **CORRECTED 2026-08-14.** The earlier wording claimed medium realism produced *significantly* higher eeriness and that trustworthiness *scaled monotonically*. Neither holds: no eeriness comparison reached significance, and only the HRA–LRA trust contrast did. Both claims now match §3.6 and §3.7.

Validated — I have read the original and agree it supports the sentence.

#46 mayer1995 — Mayer et al. (1995) *Trust and Credibility in Virtual Agents*

Sentence in the-sis: "Mayer et al. established this structure by decomposing perceived trustworthiness into ability, benevolence, and integrity, and they further separated trustworthiness as an attribute of the trustee from trust as an attitude held by the trustor."

Usage: Also cited at entry #19.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/Mayer-IntegrativeModelOrganizational-1995.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#47 mcknight2001 — McKnight & Chervany (2001) *Trust and Credibility in Virtual Agents*

Sentence in the-sis: "McKnight and Chervany subsequently carried that distinction into a typology separating trusting beliefs from trusting intentions in technology-mediated contexts."

Usage: Also cited at entry #20.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/trust/What Trust Means in E-Commerce Customer Relationships.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#48 alipour2025 — Alipour et al. (2025) *Trust and Credibility in Virtual Agents*

Sentence in the-sis: "Alipour et al. conducted a systematic review of 53 empirical studies to determine whether and how the uncanny valley effect influences trust."

Usage: First use of this source.

Original source: "The results of our systematic review reveal that most studies rely on static images or hypothetical scenarios with limited real-time interaction, and the majority use subjective trust measures."

How to check: Read directly from `Articles/alipour Would you rely on an eerie agent?.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#49 nowak2003 — Nowak & Biocca (2003) *Trust and Credibility in Virtual Agents*

Sentence in thesis: "Very high levels of this cue do not consistently produce higher social presence."

Usage: First use of this source.

Original source: "Participants interacting with the less-anthropomorphic image reported more copresence and social presence than those interacting with partners represented by either no image at all or by a highly anthropomorphic image of the other, indicating that the more anthropomorphic images set up higher expectations that lead to reduced presence when these expectations were not met."

How to check: Read directly from `Articles/Nowak2003.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#50 mori1970 — Mori (1970) *Trust and Credibility in Virtual Agents*

Sentence in thesis: "This pattern is structurally analogous to the uncanny valley, where affinity likewise ceases to scale linearly with human resemblance and inverts before recovering at full fidelity."

Usage: Also cited at entry #4.

Original source: "p. 1 ("Valley of familiarity"): "I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. I call this relation the 'uncanny valley.'" Figure 1 (the familiarity-vs-human-likeness graph, x-axis 0–100% human likeness): the curve rises from industrial robot through humanoid robot/stuffed animal, drops sharply through the trough (labelled "corpse" and "prosthetic hand," positioned close to the 100% end of the axis), then rises again to "healthy person" at 100% human likeness, the highest point on the graph (the "second peak"). p. 2 ("The significance of the uncanny"): "In Figure 1, a healthy person is at the top of the second peak.""

How to check: Same source as entry #4.

Validated — I have read the original and agree it supports the sentence.

#51 nowak2005 — Nowak & Rauh (2005) *Trust and Credibility in Virtual Agents*

Sentence in thesis: "Avatar design choices therefore function as meaningful social signals that shape participants' willingness to engage and trust."

Usage: First use of this source.

Original source: "p.158: "Participants ($N = 255$) were recruited...". Table 3 (p.170): anthropomorphism correlated with credibility, $r = .46$, $p < .01$. p.172 (Discussion): "A gendered avatar (whether masculine or feminine) was more credible than an androgynous avatar and more likely to be selected." p.174 (Conclusion): "Avatars that were more anthropomorphic were perceived to be more attractive and credible, and people were more likely to choose to be represented by them.""

How to check: Read directly from `Articles/Nowak Rauh 2005.pdf`. Confirm this passage supports the sentence above.

✔ **Validated** — I have read the original and agree it supports the sentence.

#52 canales2024 — Canales et al. (2024) *Trust and Credibility in Virtual Agents (figure caption)*

Sentence in thesis: "The three avatar stylisation levels used by Canales et al.: realistic (AF-Real), semi-realistic (AM-Mid), and caricatured (CM-Caricature)."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#53 alipour2025 — Alipour et al. (2025) *Trust and Credibility in Virtual Agents*

Sentence in thesis: "The missing category is a direct rating of trust itself."

Usage: Also cited at entry #48.

Original source: "The results of our systematic review reveal that most studies rely on static images or hypothetical scenarios with limited real-time interaction, and the majority use subjective trust measures."

How to check: Same source as entry #48.

Validated — I have read the original and agree it supports the sentence.

#54 canales2024 — Canales et al. (2024) *Trust and Credibility in Virtual Agents*

Sentence in the-sis: "However, the semi-realistic condition attracted the highest rate of be-havioural selection."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#55 gao2025 — Gao et al. (2025) *Trust and Credibility in Virtual Agents*

Sentence in the-sis: "Gao et al. found comparable self-reported trust between stylised and realistic virtual agents."

Usage: First use of this source.

Original source: "**Passage not copied yet.** The source is available but the exact sup-porting wording has not been quoted here."

How to check: Open `Articles/IEEE_VR25_TVCG.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#56 alimardani2024 — Alimardani et al. (2024) *Trust and Credibility in Virtual Agents*

Sentence in thesis: "Gao et al. found comparable self-reported trust between stylised and realistic virtual agents, and Alimardani et al. found no significant change in trust even when appearance and voice were manipulated independently."

Usage: First use of this source.

Original source: ""Trusting behaviour: No significant main or interaction effect of appearance or voice on advice-following rate. All four conditions produced statistically equivalent trust behaviour. Eeriness: The Humanlike/Robotic mismatch produced the highest eeriness ratings." (Article summary; DOI: 10.1145/3652988.3673970.)"

How to check: Read directly from `Articles/Mardani Effect of a Virtual Agent's Appearance and Voice on UncannyValley .pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#57 dubosc2025 — Dubosc et al. (2025) *Trust and Credibility in Virtual Agents*

Sentence in thesis: "Trustworthiness also trended higher for the stylised avatars, though this difference did not reach statistical significance."

Usage: First use of this source.

Original source: ""Eeriness: Significant main effect of stylisation ($F = 11.24$, $p < .001$, $\eta_p^2 = .23$). Stylised and semi-realistic produced significantly lower eeriness than realistic. Trustworthiness: Significant main effect ($F = 6.94$, $p < .001$, $\eta_p^2 = .16$). Stylised and semi-realistic were rated significantly more trustworthy than realistic. No interaction." (Article summary; DOI: 10.1007/s10055-025-01238-6.)"

How to check: Read directly from `Articles/Effect_of_avatar_stylization_and_facial_expression.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#58 cihodaru2025 — Cihodaru-Ştefanache & Podina (2025) *Trust and Credibility in Virtual Agents*

Sentence in thesis: "They identified consistent stylisation across all visual features as the most reliably recommended design strategy for avoiding uncanny valley responses, alongside accurate lip synchronisation and consistency between visual and voice realism levels."

Usage: First use of this source.

Original source: ""Stylisation: studies comparing stylised versus realistic ECAs consistently found that stylised agents produced lower uncanniness scores. Deliberate low-realism stylisation is the most frequently recommended avoidance strategy. Physical appearance: use consistent realism across all visual features; avoid partial stylisation." (Article summary; DOI: 10.3389/fpsyg.2025.1625984.)"

How to check: Read directly from `Articles/The_uncanny_valley_effect_in_embodied_conversation.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#59 bartneck2009 — Bartneck et al. (2009) *Trust and Credibility in Virtual Agents*

Sentence in thesis: "The Godspeed Questionnaire Series, developed by Bartneck et al., offers one such option, providing validated semantic differential scales for five perceptual dimensions."

Usage: Also cited at entry #21.

Original source: "[Abstract:] "The results have been distilled into five consistent questionnaires using semantic differential scales." [§2:] "In semantic differential scales the respondent is asked to indicate his or her position on a scale between two bipolar words, the anchors. . . a semantic differential format may effectively reduce acquiescence bias without lowering psychometric quality." Figure 1 shows a five point scale ("Strong 1 2 3 4 5 Weak"). Item counts per index are as reported by Ho & MacDorman (2010, §2.1.2), who used the instrument directly: "composed of five indices and 24 semantic differential items. The anthropomorphism index has five items, the animacy index has six items, the likeability index has five items, the perceived intelligence index has five items, and the perceived safety index has three items.""

How to check: Same source as entry #21.

Validated — I have read the original and agree it supports the sentence.

#60 macdorman2006 — MacDorman & Ishiguro (2006) *Trust and Credibility in Virtual Agents*

Sentence in the-sis: "The eeriness-specific instruments introduced by MacDorman and Ishiguro complement the Godspeed series by targeting the negative affective response to near-human imperfection directly."

Usage: Also cited at entry #37.

Original source: "p. 297 abstract: "The development of robots that closely resemble human beings can contribute to cognitive research. An android provides an experimental apparatus that has the potential to be controlled more precisely than any human actor." p. 298: "An android is defined to be 'an artificial system designed with the ultimate goal of being indistinguishable from humans in its external appearance and behavior' (MacDorman & Ishiguro, 2005).""

How to check: Same source as entry #37.

Validated — I have read the original and agree it supports the sentence.

#61 ho2010 — Ho & MacDorman (2010) *Trust and Credibility in Virtual Agents*

Sentence in the-sis: "Ho and MacDorman later refined this into a two-subscale instrument that distinguishes perceived eeriness from perceived human likeness, and their validation established that the two dimensions are orthogonal."

Usage: Also cited at entry #22.

Original source: "[Abstract:] "Phase 2 of this study developed alternative indices with non-significant correlations ($p > .05$) between the proposed y-axis eeriness and x-axis perceived humanness ($r = .02$)." [§3.3.2 Eeriness index:] "Therefore, the final version of the eeriness index would retain eight items that explained 62.04% of the variance and held an overall internal reliability of .74." [§3.3.1 Humanness index:] "Therefore, the final version of the humanness index would retain six items.""

How to check: Same source as entry #22.

Validated — I have read the original and agree it supports the sentence.

#62 kumar2019 — Kumar & Poornima (2019) *NPR for Virtual Humans*

Sentence in the-sis: "Kumar and Poornima classified these techniques into four categories."

Usage: First use of this source.

Original source: ""Taxonomy: Stroke-based rendering, Colour image analogy, Region-based rendering, Image filtering abstraction, Stylisation... covers NPR techniques from 1963 to 2017." (Article summary; DOI: 10.1007/s42044-019-00034-1.)"

How to check: Read directly from `Articles/NPR/Kumar- A comprehensive survey on NPR.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#63 kumar2019 — Kumar & Poornima (2019) *NPR for Virtual Humans (figure caption)*

Sentence in the-sis: "Kumar and Poornima's classification of NPR techniques into four families: stroke-based rendering, image analogy, region-based techniques, and image filtering."

Usage: Also cited at entry #62.

Original source: ""Taxonomy: Stroke-based rendering, Colour image analogy, Region-based rendering, Image filtering abstraction, Stylisation... covers NPR techniques from 1963 to 2017." (Article summary; DOI: 10.1007/s42044-019-00034-1.)"

How to check: Same source as entry #62.

Validated — I have read the original and agree it supports the sentence.

#64 lake2000 — Lake et al. (2000) *NPR for Virtual Humans (figure caption)*

Sentence in the-sis: "A cartoon character rendered with toon shading, showing flat tonal bands separated by hard edges."

Usage: First use of this source.

Original source: "[§3.1 "Cartoon Shading":] a 1D lighting value indexes a small (1–2 texel) gradient texture, producing a small number of discrete, hard-edged tonal bands rather than a continuous Lambertian ramp."

How to check: Read directly from `Articles/NPR/npar2000_lake_et_al.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#65 lake2000 — Lake et al. (2000) *NPR for Virtual Humans*

Sentence in the-sis: "It quantises the continuous diffuse lighting term into a small number of discrete bands."

Usage: Also cited at entry #64.

Original source: "[§3.1 "Cartoon Shading":] a 1D lighting value indexes a small (1–2 texel) gradient texture, producing a small number of discrete, hard-edged tonal bands rather than a continuous Lambertian ramp."

How to check: Same source as entry #64.

Validated — I have read the original and agree it supports the sentence.

#66 barla2006 — Barla et al. (2006) *NPR for Virtual Humans*

Sentence in the-sis: "extended toon shading's quantisation into a richer parameter space with X-Toon."

Usage: First use of this source.

Original source: "In this paper we describe X-Toon, a toon shader that supports view-dependent effects through two independent extensions to conventional toon shading. The first incorporates a notion of tone detail, so that tone varies with depth or orientation relative to the camera. For this, we replace the conventional 1D texture used in toon shading with a 2D texture, where the second dimension corresponds to desired tone 'detail'."

How to check: Read directly from `Articles/NPR/x-toon.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#67 raskar1999 — Raskar & Cohen (1999) *NPR for Virtual Humans*

Sentence in the-sis: "Raskar and Cohen proposed processing the mesh twice from the same viewpoint."

Usage: First use of this source.

Original source: "[§3.2 "Translated Back-facing Polygons":] front-facing polygons are rendered normally, then back-facing polygons are rendered a second time, translated toward the viewer by a fixed amount t or a depth-scaled amount $k \cdot z$, so the translated back-facing geometry appears in front of the front-facing surface along silhouette boundaries."

How to check: Read directly from `Articles/Raskar -Image Precision Silhouette Edges.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#68 isenberg2003 — Isenberg et al. (2003) *NPR for Virtual Humans*

Sentence in the-sis: "They distinguish silhouette edges from feature lines such as creases, borders, and self-intersections."

Usage: Also cited at entry #9.

Original source: "[Classification section, p. 29:] "we distinguish between image space algorithms that only operate on image buffers, hybrid algorithms that perform manipulations in object space but yield the silhouette in an image buffer, and object space algorithms that perform all calculations in object space." [Image space algorithms section, p. 29:] "Image space algorithms exploit discontinuities in the image buffer(s) that result from conventional rendering and extract them using image-processing methods.""

How to check: Same source as entry #9.

Validated — I have read the original and agree it supports the sentence.

#69 gonzalez2018 — Gonzalez & Woods (2018) *NPR for Virtual Humans*

Sentence in the-sis: "The Sobel operator, a discrete gradient approximation applied to the diffuse texture via a 3×3 convolution kernel, identifies luminance boundaries corresponding to clothing patterns, facial contours, and painted surface details."

Usage: Also cited at entry #12.

Original source: "[p. 18, §1.1 "What Is Digital Image Processing?":] "Note that a digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels, and pixels." [p. 66, Ch. 2, Eq. (2-9)–(2-10):] the image is formally represented as an $M \times N$ array $f(x, y)$, and "[e]ach element of this array is called an image element, picture element, pixel, or pel." [p. 702, §10.2 "Point, Line, and Edge Detection":] "Edge pixels are pixels at which the intensity of an image changes abruptly, and edges (or edge segments) are sets of connected edge pixels.""

How to check: Same source as entry #12.

Validated — I have read the original and agree it supports the sentence.

#70 marr1980 — Marr & Hildreth (1980) *NPR for Virtual Humans*

Sentence in the-sis: "Marr and Hildreth demonstrated that low-pass prefiltering is a prerequisite for stable gradient computation."

Usage: Also cited at entry #29.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: D. Marr and E. Hildreth, "Theory of Edge Detection," *Proc. R. Soc. Lond. B* 207, 1980, pp. 187–217, DOI 10.1098/rspb.1980.0020"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#71 kyprianidis2009 — Kyprianidis et al. (2009) *NPR for Virtual Humans*

Sentence in the-sis: "This yields brush-stroke effects oriented along salient contours."

Usage: Also cited at entry #32.

Original source: "[Abstract:] "We present a non-photorealistic rendering technique to transform color images and videos into painterly abstractions. It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor." [§3.1 "Orientation and Anisotropy Estimation":] "We calculate the structure tensor directly from the RGB values of the input. . . The structure tensor of f is then defined as" the matrix of pairwise dot products $f_x \cdot f_x$, $f_x \cdot f_y$, $f_y \cdot f_y$, subsequently smoothed with a Gaussian; "Local orientation and a measure for the anisotropy are then derived from the eigenvalues and eigenvectors of the smoothed structure tensor.""

How to check: Same source as entry #32.

Validated — I have read the original and agree it supports the sentence.

#72 kyprianidis2011 — Kyprianidis (2011) *NPR for Virtual Humans*

Sentence in the-sis: "A multiscale variant of this filter, also by Kyprianidis, propagates orientation estimates and filter outputs through a resolution pyramid."

Usage: First use of this source.

Original source: ""A multi-scale computation scheme is proposed that simultaneously propagates local orientation estimates and filtering results up a low-pass filtered pyramid. This allows for a strong abstraction effect and avoids artifacts in large low-contrast regions." (Physical PDF confirmed: anisotropic_multicale_209.pdf — NPAR 2011, p. 55.)"

How to check: Read directly from `Articles/NPR/anisotropic_multicale_209.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#73 kyprianidis2011 — Kyprianidis (2011) *NPR for Virtual Humans (figure caption)*

Sentence in the-sis: "Painterly abstraction produced by Kuwahara filtering. The proposed method in panel (c) is the multiscale variant, which suppresses the artefacts the anisotropic filter leaves in low-contrast regions."

Usage: Also cited at entry #72.

Original source: ""A multi-scale computation scheme is proposed that simultaneously propagates local orientation estimates and filtering results up a low-pass filtered pyramid. This allows for a strong abstraction effect and avoids artifacts in large low-contrast regions." (Physical PDF confirmed: anisotropic_multicale_209.pdf — NPAR 2011, p. 55.)"

How to check: Same source as entry #72.

Validated — I have read the original and agree it supports the sentence.

#74 praun2001 — Praun et al. (2001) *NPR for Virtual Humans*

Sentence in the-sis: "demonstrated real-time hatching as a complementary branch of this design space."

Usage: First use of this source.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/NPR/hatching.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#75 wisessing2016 — Wisessing et al. (2016) *NPR for Virtual Humans*

Sentence in the-sis: "Wisessing et al. demonstrated that toon shading reduces the perceived intensity of facial emotional expression while leaving recognition accuracy intact."

Usage: First use of this source.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: P. Wisessing et al., "Perception of Lighting and Shading for Animated Virtual Characters," *ACM SAP* 2016"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#76 wisessing2020 — Wisessing et al. (2020) *NPR for Virtual Humans*

Sentence in the-sis: "Higher brightness made characters more appealing regardless of the emotion portrayed. Darker shadows raised eeriness and reduced the appeal of cartoon characters, while leaving the appeal of realistic characters unaffected."

Usage: First use of this source.

Original source: "**Primary source read** (`Articles/Wissessing-Enloghten me.pdf`). Main findings: "Higher brightness makes for more appealing characters, regardless of the emotion they portray" and "Darker shadows are unappealing for cartoon characters but do not affect the appeal of realistic characters." Results: "A main effect of KTFR on eeriness showed 16:1 was more eerie than 4:1 or 1:1 ($p < 0.0276$ in both cases)."""

How to check: **CORRECTED 2026-08-14.** The earlier wording claimed brighter light and lighter shadows reduced eeriness *for both* photorealistic and stylised characters, and that the reduction was larger for photorealistic rendering from a higher baseline. The brightness–eeriness effect was a three-way interaction confined to male Toon characters, the shadow effect came mainly from the male character, and the higher-baseline claim has no support in the paper. The appeal claim was also inverted for realistic characters.

Validated — I have read the original and agree it supports the sentence.

#77 zibrek2019tap — Zibrek et al. (2019) *NPR for Virtual Humans*

Sentence in the-sis: "Zibrek et al. conducted a within-participants VR study in which the same animated character was presented in photorealistic and non-photorealistic render conditions."

Usage: First use of this source.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: K. Zibrek, S. Martin, R. McDonnell, "Is Photorealism Important for Perception of Expressive Virtual Humans in VR?," *ACM TAP* 16(3), 2019"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#78 zibrek2019mig — Zibrek & McDonnell (2019) *NPR for Virtual Humans*

Sentence in the-sis: "Zibrek and McDonnell extended this investigation to embodied VR. Photorealism significantly increased both place illusion, the sense of physically inhabiting the virtual space, and self-reported social presence. The behavioural measure of interpersonal proximity was unaffected by render style."

Usage: First use of this source.

Original source: "**Primary source read** (`Articles/02_trust_and_avatar_perception/3359566.3360064.pdf`). Results §2.3: "We found a significant difference in Social Presence, where participants felt more present with the Realistic virtual character than the Simple one ($t(18) = 2.45$, $p = 0.025$, $d = 0.56$)." Results §2.4: "Wilcoxon Matched Pairs Test showed a significant difference between the Realistic room and Simple room ($Z = 2.02$, $p = 0.04$, $r = 0.46$), where the realistic environment increased the place illusion." Introduction: "Our previous study [Zibrek et al. 2019] . . . found that place illusion was higher in the photorealistic condition, while social presence and empathetic concern were not affected. Our new results show an increase in self-reported social presence in the photorealistic condition.""

How to check: **CONFIRMED (corrected 2026-08-14).** The earlier wording claimed a "weaker and less consistent effect on social presence," which contradicted the source: social presence was significantly higher for the realistic character ($p = 0.025$, $d = 0.56$). That no-effect result belongs to `zibrek2019tap`, against which this paper explicitly contrasts itself, and the two Zibrek findings had been conflated. The thesis sentence now states both significant self-report increases and the unaffected behavioural proximity measure, matching the source.

Validated — I have read the original and agree it supports the sentence.

#79 zibrek2019mig — Zibrek & McDonnell (2019) *Evaluation §6.3.5 The Video-Based Setting*

Sentence in the-sis: "A comparable VR study held shape and animation constant and varied only render style. Photorealism significantly raised both place illusion and social presence."

Usage: Second use of this source. Supports the claim that a rendering difference undetectable on flat video may register in an embodied setting.

Original source: "**Primary source read.** Method: "We created two render styles of the environment and character: realistic . . . and simple (where photorealistic render effects were removed), while keeping the shape and animation the same." Results as quoted in the entry above ($p = 0.025$, $d = 0.56$ for social presence; $p = 0.04$, $r = 0.46$ for place illusion)."

How to check: **CONFIRMED.** Both the design description (shape and animation held constant) and both significance claims match the source exactly.

Validated — I have read the original and agree it supports the sentence.

#80 latoschik2017 — Latoschik et al. (2017) *NPR for Virtual Humans*

Sentence in the-sis: "They compared an abstract wooden mannequin against high fidelity avatars generated from photogrammetric scans. The realistic avatars were rated significantly more human-like and evoked stronger virtual body ownership, alongside some indication of an uncanny valley response."

Usage: First use of this source.

Original source: "**Primary source read** (`Articles/Latochik- the imapct.pdf.pdf`). Abstract: "We compared abstract avatar representations based on a wooden mannequin with high fidelity avatars generated from photogrammetry 3D scan methods . . . The realistic avatars were rated significantly more human-like when used as avatars for the others and evoked a stronger acceptance in terms of virtual body ownership (VBO). There also was some indication of a potential uncanny valley.""

How to check: **CORRECTED 2026-08-14.** The earlier wording described *three* conditions (abstract, semi-realistic, realistic) and attributed findings to a semi-realistic condition. The study has only *two* conditions and no semi-realistic level exists. The downstream conclusion about a "threshold of representational fidelity" rested entirely on that non-existent condition and has been replaced.

Validated — I have read the original and agree it supports the sentence.

#81 canales2024 — Canales et al. (2024) *NPR for Virtual Humans*

Sentence in the-sis: "The most directly related prior work on the avatar side, Canales et al., applied stylisation to custom assets in an open pipeline and compared coarse-grained style levels rather than specific technical approaches."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#82 roshaan2026 — Roshaan (2026) *NPR for Virtual Humans*

Sentence in thesis: "Roshaan combined depth, normal and colour cues into a silhouette, structure and texture hierarchy, merged through max-pooling and scaled by scene luminance for consistent line quality without manual retuning."

Usage: Also cited at entries #20, #118 (Methodology/Implementation, V5).

Original source: "Abstract: "a unified, real-time post-processing shader architecture that makes use of adaptive sensitivity and a three-layer hierarchy... breaking [the scene] down into separate Silhouette, Structure and Texture layers... a Max-pooling fusion technique." Contributions (§Adaptability): "adaptive sensitivity mechanism... dynamically modulated the edge strength based on scene luminance and depth.""

How to check: Primary source read directly (Articles/03_npr_rendering/Adaptive Hierarchical Edge Detection.pdf, title page + p. 1). CONFIRMED.

Validated — I have read the original and agree it supports the sentence.

#83 mcdonnell2012 — McDonnell et al. (2012) *Research Gap*

Sentence in thesis: "and Zibrek et al."

Usage: Also cited at entry #2.

Original source: "§4: "We created 11 rendering styles in total for use in our experiments." §5.1.3 Realism: "Posthoc analysis showed that six distinct levels of realism were found" (this is the source of the previous "six" error — six is the number of statistically grouped realism levels, not the number of styles tested). Appeal: "four of the most abstract and the two most realistic styles were rated as the most appealing... and significantly more appealing than all other styles... characters that were rated halfway between abstract and realistic... were considered the least appealing." Re-assuring: "the most realistic and most abstract styles were considered the most re-assuring... HumanIll was rated as the most eerie." Trustworthy: "the most realistic Humans, and all other Toons were rated as equally trustworthy, and more trustworthy than all other styles." §7 Discussion: "These results are very much in line with the Uncanny Valley theory. Contrary to the theory, however, was the fact that our most realistic character was often rated as appealing or pleasant as our cartoon characters. The drop in appeal for our stimuli came from characters that were rated as unfamiliar or those that were in the middle of our 'abstract to realistic' scale.""

How to check: Same source as entry #2.

Validated — I have read the original and agree it supports the sentence.

#84 zibrek2019tap — Zibrek et al. (2019) *Research Gap*

Sentence in the-sis: "and Zibrek et al."

Usage: Also cited at entry #77.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: K. Zibrek, S. Martin, R. McDonnell, "Is Photorealism Important for Perception of Expressive Virtual Humans in VR?," *ACM TAP* 16(3), 2019"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#85 canales2024 — Canales et al. (2024) *Research Gap*

Sentence in the-sis: "and Gao et al."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#86 gao2025 — Gao et al. (2025) *Research Gap*

Sentence in the-sis: "and Gao et al."

Usage: Also cited at entry #55.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/IEEE_VR25_TVCG.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#87 canales2024 — Canales et al. (2024) *Research Gap*

Sentence in the-sis: "is the closest prior work, but compared coarse style categories rather than a specific NPR shader pipeline, used a between-subjects design, and did not measure eeriness or anthropomorphism as separate dimen-sions."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#88 bartneck2009 — Bartneck et al. (2009) *Research Gap*

Sentence in the-sis: "It adopts the likeability subscale of the Godspeed questionnaire."

Usage: Also cited at entry #21.

Original source: "[Abstract:] "The results have been distilled into five consistent question-naires using semantic differential scales." [§2:] "In semantic differential scales the respondent is asked to indicate his or her position on a scale between two bipolar words, the anchors. . . a semantic differential format may effectively reduce acquiescence bias without lowering psychometric quality." Figure 1 shows a five point scale ("Strong 1 2 3 4 5 Weak"). Item counts per index are as reported by Ho & MacDorman (2010, §2.1.2), who used the instrument directly: "composed of five indices and 24 semantic differential items. The anthropomorphism index has five items, the animacy index has six items, the likeability index has five items, the perceived intelligence index has five items, and the perceived safety index has three items.""

How to check: Same source as entry #21.

Validated — I have read the original and agree it supports the sentence.

#89 macdorman2006 — MacDorman & Ishiguro (2006) *Research Gap*

Sentence in the-sis: "Eeriness is measured using scales grounded in MacDorman and Ishiguro and refined by Ho and MacDorman."

Usage: Also cited at entry #37.

Original source: "p. 297 abstract: "The development of robots that closely resemble human beings can contribute to cognitive research. An android provides an experimental apparatus that has the potential to be controlled more precisely than any human actor." p. 298: "An android is defined to be 'an artificial system designed with the ultimate goal of being indistinguishable from humans in its external appearance and behavior' (MacDorman & Ishiguro, 2005).""

How to check: Same source as entry #37.

Validated — I have read the original and agree it supports the sentence.

#90 ho2010 — Ho & MacDorman (2010) *Research Gap*

Sentence in the-sis: "Eeriness is measured using scales grounded in MacDorman and Ishiguro and refined by Ho and MacDorman."

Usage: Also cited at entry #22.

Original source: "[Abstract:] "Phase 2 of this study developed alternative indices with non-significant correlations ($p > .05$) between the proposed y-axis eeriness and x-axis perceived humanness ($r = .02$)." [§3.3.2 Eeriness index:] "Therefore, the final version of the eeriness index would retain eight items that explained 62.04% of the variance and held an overall internal reliability of .74." [§3.3.1 Humanness index:] "Therefore, the final version of the humanness index would retain six items.""

How to check: Same source as entry #22.

Validated — I have read the original and agree it supports the sentence.

#91 mayer1995 — Mayer et al. (1995) *Research Gap*

Sentence in the-sis: "A multi-component trust battery is structured around the ability, benevolence, and integrity dimensions of the Mayer et al. integrative trust model, operationalised within the trusting-beliefs typology of McKnight and Chervany."

Usage: Also cited at entry #19.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/Mayer-IntegrativeModelOrganizational-1995.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#92 mcknight2001 — McKnight & Chervany (2001) *Research Gap*

Sentence in the-sis: "A multi-component trust battery is structured around the ability, benev-olence, and integrity dimensions of the Mayer et al. integrative trust model, operationalised within the trusting-beliefs typology of McKnight and Chervany."

Usage: Also cited at entry #20.

Original source: "**Passage not copied yet.** The source is available but the exact sup-porting wording has not been quoted here."

How to check: Open `Articles/trust/What Trust Means in E-Commerce Customer Relationships.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#93 canales2024 — Canales et al. (2024) *Research Gap*

Sentence in the-sis: "A forced-choice behavioural preference task supplements the self-report battery, following the paradigm of Canales et al. and the methodological prescription of Alipour et al."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#94 alipour2025 — Alipour et al. (2025) *Research Gap*

Sentence in the-sis: "A forced-choice behavioural preference task supplements the self-report battery, following the paradigm of Canales et al. and the methodological prescription of Alipour et al."

Usage: Also cited at entry #48.

Original source: "The results of our systematic review reveal that most studies rely on static images or hypothetical scenarios with limited real-time interaction, and the majority use subjective trust measures."

How to check: Same source as entry #48.

Validated — I have read the original and agree it supports the sentence.

Chapter: Methodology

#95 mori1970 — Mori (1970) *Research Questions*

Sentence in thesis: "Uncanny valley theory places a real human speaker at the peak of the affinity curve, since maximal human likeness corresponds to maximal comfort."

Usage: Also cited at entry #4.

Original source: "p. 1 ("Valley of familiarity"): "I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. I call this relation the 'uncanny valley.'" Figure 1 (the familiarity-vs-human-likeness graph, x-axis 0–100% human likeness): the curve rises from industrial robot through humanoid robot/stuffed animal, drops sharply through the trough (labelled "corpse" and "prosthetic hand," positioned close to the 100% end of the axis), then rises again to "healthy person" at 100% human likeness, the highest point on the graph (the "second peak"). p. 2 ("The significance of the uncanny"): "In Figure 1, a healthy person is at the top of the second peak.""

How to check: Same source as entry #4.

Validated — I have read the original and agree it supports the sentence.

#96 mixamo — Adobe Inc. (2024) *Avatar Platforms and Rendering Constraints*

Sentence in thesis: "The Mixamo platform provides an online library of rigged humanoid characters and skinned animations that can be exported to Unity as standard FBX assets."

Usage: First use of this source.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open mixamo.com and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#97 metasdk — Meta Platforms (2024) *Avatar Platforms and Rendering Constraints*

Sentence in the-sis: "Avatar rendering, face tracking, and body tracking are handled internally by the Meta Avatars SDK, Meta's avatar framework for the Meta Quest 3."

Usage: First use of this source.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open Meta Avatars SDK documentation and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#98 avaturn — Avaturn (2024) *Avatar Platforms and Rendering Constraints*

Sentence in the-sis: "Avaturn is an avatar creation service that reconstructs a personalised rigged avatar with facial blendshape targets from three uploaded photographs (Figure *[cross-ref]*)."

Usage: First use of this source.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open docs.avaturn.me and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#99 gltf2 — Khronos Group (2017) *Avatar Platforms and Rendering Constraints*

Sentence in the-sis: "The service exports its output as `.glb` files, which implement the glTF 2.0 standard and bundle geometry, PBR materials, and skeletal rigging within a single self-contained file."

Usage: First use of this source.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open Khronos glTF 2.0 specification and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#100 isenberg2003 — Isenberg et al. (2003) *NPR Technique Design*

Sentence in the-sis: "NPR techniques in computer graphics encompass a wide range of abstraction levels."

Usage: Also cited at entry #9.

Original source: "[Classification section, p. 29:] "we distinguish between image space algorithms that only operate on image buffers, hybrid algorithms that perform manipulations in object space but yield the silhouette in an image buffer, and object space algorithms that perform all calculations in object space." [Image space algorithms section, p. 29:] "Image space algorithms exploit discontinuities in the image buffer(s) that result from conventional rendering and extract them using image-processing methods.""

How to check: Same source as entry #9.

Validated — I have read the original and agree it supports the sentence.

#101 zell2015 — Zell et al. (2015) *NPR Technique Design*

Sentence in the-sis: "Choosing a level that is too low may place an avatar within the uncanny valley, whereas excessive stylisation can reduce visual detail and compromise expressive readability."

Usage: Also cited at entry #10.

Original source: "Main Findings list, p. 184:2: "The perceived intensity of expressions decreases with realism of shape, but is nearly independent of material stylization.""

How to check: Same source as entry #10.

Validated — I have read the original and agree it supports the sentence.

#102 wallraven2008 — Wallraven et al. (2008) *NPR Technique Design*

Sentence in the-sis: "Choosing a level that is too low may place an avatar within the uncanny valley, whereas excessive stylisation can reduce visual detail and compromise expressive readability."

Usage: Also cited at entry #11.

Original source: "§4 Summary, p. 23:19: "In addition, in Experiment 2, we found clear effects of shape blurring on both intensity and sincerity judgments. This demonstrates that, even though reduced shape information does not affect recognition, it does influence the perceived quality of the expression.""

How to check: Same source as entry #11.

Validated — I have read the original and agree it supports the sentence.

#103 isenberg2003 — Isenberg et al. (2003) *NPR Technique Design → V1 — Inverted Hull Outline*

Sentence in the-sis: "Object silhouettes are a defining visual feature of nonphotorealistic rendering."

Usage: Also cited at entry #9.

Original source: "[Classification section, p. 29:] "we distinguish between image space algorithms that only operate on image buffers, hybrid algorithms that perform manipulations in object space but yield the silhouette in an image buffer, and object space algorithms that perform all calculations in object space." [Image space algorithms section, p. 29:] "Image space algorithms exploit discontinuities in the image buffer(s) that result from conventional rendering and extract them using image-processing methods.""

How to check: Same source as entry #9.

Validated — I have read the original and agree it supports the sentence.

#104 isenberg2003 — Isenberg et al. (2003) *NPR Technique Design → V1 — Inverted Hull Outline*

Sentence in the-sis: "The inverted hull method, also known as the shell method, was selected because it produces stable geometric silhouettes without requiring screen-space processing."

Usage: Also cited at entry #9.

Original source: "[Classification section, p. 29:] "we distinguish between image space algorithms that only operate on image buffers, hybrid algorithms that perform manipulations in object space but yield the silhouette in an image buffer, and object space algorithms that perform all calculations in object space." [Image space algorithms section, p. 29:] "Image space algorithms exploit discontinuities in the image buffer(s) that result from conventional rendering and extract them using image-processing methods.""

How to check: Same source as entry #9.

Validated — I have read the original and agree it supports the sentence.

#105 isenberg2003 — Isenberg et al. (2003) *NPR Technique Design → V1 — Inverted Hull Outline*

Sentence in the-sis: "This approach is well suited to real-time NPR avatar rendering because it requires no offline mesh preprocessing and executes entirely within the GPU vertex transformation stage."

Usage: Also cited at entry #9.

Original source: "[Classification section, p. 29:] "we distinguish between image space algorithms that only operate on image buffers, hybrid algorithms that perform manipulations in object space but yield the silhouette in an image buffer, and object space algorithms that perform all calculations in object space." [Image space algorithms section, p. 29:] "Image space algorithms exploit discontinuities in the image buffer(s) that result from conventional rendering and extract them using image-processing methods.""

How to check: Same source as entry #9.

Validated — I have read the original and agree it supports the sentence.

#106 isenberg2003 — Isenberg et al. (2003) *NPR Technique Design → V1 — Inverted Hull Outline*

Sentence in the-sis: "Consequently, creases between body regions, clothing seams, and facial contours receive no edge marking, as the method captures only the external silhouette of the mesh."

Usage: Also cited at entry #9.

Original source: "[Classification section, p. 29:] "we distinguish between image space algorithms that only operate on image buffers, hybrid algorithms that perform manipulations in object space but yield the silhouette in an image buffer, and object space algorithms that perform all calculations in object space." [Image space algorithms section, p. 29:] "Image space algorithms exploit discontinuities in the image buffer(s) that result from conventional rendering and extract them using image-processing methods.""

How to check: Same source as entry #9.

Validated — I have read the original and agree it supports the sentence.

#107 lake2000 — Lake et al. (2000) *NPR Technique Design → V2 — Toon Shading*

Sentence in the-sis: "Toon shading, also known as cel shading, is one of the most widely adopted techniques in NPR, where stylisation is achieved by simplifying the tonal representation of a surface rather than reproducing physically accurate illumination."

Usage: Also cited at entry #64.

Original source: "[§3.1 "Cartoon Shading":] a 1D lighting value indexes a small (1–2 texel) gradient texture, producing a small number of discrete, hard-edged tonal bands rather than a continuous Lambertian ramp."

How to check: Same source as entry #64.

Validated — I have read the original and agree it supports the sentence.

#108 lake2000 — Lake et al. (2000) *NPR Technique Design → V2 — Toon Shading*

Sentence in the-sis: "Correspondingly, the technique is well suited to interactive character rendering."

Usage: Also cited at entry #64.

Original source: "[§3.1 "Cartoon Shading":] a 1D lighting value indexes a small (1–2 texel) gradient texture, producing a small number of discrete, hard-edged tonal bands rather than a continuous Lambertian ramp."

How to check: Same source as entry #64.

Validated — I have read the original and agree it supports the sentence.

#109 lake2000 — Lake et al. (2000) *NPR Technique Design → V2 — Toon Shading*

Sentence in the-sis: "This operation replaces continuous illumination gradients with discrete tonal transitions, producing the characteristic stepped appearance of toon shading."

Usage: Also cited at entry #64.

Original source: "[§3.1 "Cartoon Shading":] a 1D lighting value indexes a small (1–2 texel) gradient texture, producing a small number of discrete, hard-edged tonal bands rather than a continuous Lambertian ramp."

How to check: Same source as entry #64.

Validated — I have read the original and agree it supports the sentence.

#110 barla2006 — Barla et al. (2006) *NPR Technique Design → V2.2 — X-Toon Shading*

Sentence in thesis: "X-Toon shading extends conventional toon shading by replacing the 1D toon ramp with a 2D ramp texture, where the horizontal axis represents the lighting response and the vertical axis controls the abstraction level."

Usage: Also cited at entry #66.

Original source: "In this paper we describe X-Toon, a toon shader that supports view-dependent effects through two independent extensions to conventional toon shading. The first incorporates a notion of tone detail, so that tone varies with depth or orientation relative to the camera. For this, we replace the conventional 1D texture used in toon shading with a 2D texture, where the second dimension corresponds to desired tone 'detail'."

How to check: Same source as entry #66.

Validated — I have read the original and agree it supports the sentence.

#111 barla2006 — Barla et al. (2006) *NPR Technique Design → V2.2 — X-Toon Shading*

Sentence in thesis: "In addition to tonal abstraction, Barla et al. described a shape detail extension that modifies the surface normal field by blending the original mesh normals with normals derived from a geometrically simplified shape."

Usage: Also cited at entry #66.

Original source: "In this paper we describe X-Toon, a toon shader that supports view-dependent effects through two independent extensions to conventional toon shading. The first incorporates a notion of tone detail, so that tone varies with depth or orientation relative to the camera. For this, we replace the conventional 1D texture used in toon shading with a 2D texture, where the second dimension corresponds to desired tone 'detail'."

How to check: Same source as entry #66.

Validated — I have read the original and agree it supports the sentence.

#112 barla2006 — Barla et al. (2006) *NPR Technique Design → V2.2 — X-Toon Shading*

Sentence in the-sis: "Barla et al. further described an attribute mapping mechanism that controls the abstraction level through three driving modes: depth, curvature and manually controlled."

Usage: Also cited at entry #66.

Original source: "In this paper we describe X-Toon, a toon shader that supports view-dependent effects through two independent extensions to conventional toon shading. The first incorporates a notion of tone detail, so that tone varies with depth or orientation relative to the camera. For this, we replace the conventional 1D texture used in toon shading with a 2D texture, where the second dimension corresponds to desired tone 'detail'."

How to check: Same source as entry #66.

Validated — I have read the original and agree it supports the sentence.

#113 isenberg2003 — Isenberg et al. (2003) *NPR Technique Design → V3 — Screen Space Normal Edge Detection*

Sentence in the-sis: "Silhouette extraction techniques can be classified into image space, object space, and hybrid approaches."

Usage: Also cited at entry #9.

Original source: "[Classification section, p. 29:] "we distinguish between image space algorithms that only operate on image buffers, hybrid algorithms that perform manipulations in object space but yield the silhouette in an image buffer, and object space algorithms that perform all calculations in object space." [Image space algorithms section, p. 29:] "Image space algorithms exploit discontinuities in the image buffer(s) that result from conventional rendering and extract them using image-processing methods.""

How to check: Same source as entry #9.

Validated — I have read the original and agree it supports the sentence.

#114 saitotakahashi1990 — Saito & Takahashi (1990) *NPR Technique Design → V3 — Screen Space Normal Edge Detection*

Sentence in thesis: "In prior work, per-pixel surface normals stored in G-buffer channels have been used as a cue for edge detection."

Usage: First use of this source.

Original source: "[Abstract, confirmed from SIGGRAPH History page:] "data about geometric properties of surfaces are preserved as Geometric Buffers (G-buffers), with each G-buffer containing one geometric property such as the depth or the normal vector of each pixel" and the algorithm detects "discontinuities, edges, contour lines, and curved hatching.""

How to check: Read directly from `Articles/NPR/Comprehensible Rendering of 3-D Shapes Saito.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#115 nienhaus2003 — Nienhaus & Döllner (2003) *NPR Technique Design → V3 — Screen Space Normal Edge Detection*

Sentence in thesis: "Discontinuities across these same channels detect silhouette, border, and crease edges in real-time NPR rendering, and crease edges in particular are recovered from the normal buffer."

Usage: First use of this source.

Original source: "[Abstract, p. 1:] "It is based on the edge map, a 2D texture that encodes visually important edges of 3D scene objects, and includes silhouette edges, border edges, and crease edges." [Section 3, p. 3:] "A silhouette edge is an edge adjacent to a polygon facing towards the camera (front-facing) and one polygon facing in the opposite direction (back-facing). A border edge is an edge to exactly one polygon. A crease edge is an edge between two front-facing (or back-facing, respectively) polygons whose dihedral angle is above some threshold." [Section 4.1.2, p. 3–4:] "Abrupt changes in the normal-buffer occur typically at crease edges... 0th order discontinuities of the normal-buffer indicate crease edges and, furthermore, profile edges.""

How to check: Read directly from `Articles/NPR/Edge Enhancement.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#116 gonzalez2018 — Gonzalez & Woods (2018) *NPR Technique Design → V4 — Sobel Edge Detection*

Sentence in the-sis: "The Sobel operator is a gradient-based edge detector and one of the most commonly applied methods in digital image processing."

Usage: Also cited at entry #12.

Original source: "[p. 18, §1.1 "What Is Digital Image Processing?":] "Note that a digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels, and pixels." [p. 66, Ch. 2, Eq. (2-9)–(2-10):] the image is formally represented as an $M \times N$ array $f(x, y)$, and "[e]ach element of this array is called an image element, picture element, pixel, or pel." [p. 702, §10.2 "Point, Line, and Edge Detection":] "Edge pixels are pixels at which the intensity of an image changes abruptly, and edges (or edge segments) are sets of connected edge pixels.""

How to check: Same source as entry #12.

Validated — I have read the original and agree it supports the sentence.

#117 marr1980 — Marr & Hildreth (1980) *NPR Technique Design → V4.2 — Gaussian Prefiltered Sobel*

Sentence in the-sis: "This reduces variation at fine spatial scales without the ringing artefacts produced by box-averaging filters."

Usage: Also cited at entry #29.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: D. Marr and E. Hildreth, "Theory of Edge Detection," *Proc. R. Soc. Lond. B* 207, 1980, pp. 187–217, DOI 10.1098/rspb.1980.0020"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#118 gonzalez2018 — Gonzalez & Woods (2018) *NPR Technique Design → V4.2 — Gaussian Prefiltered Sobel*

Sentence in thesis: "This reduces variation at fine spatial scales without the ringing artefacts produced by box-averaging filters."

Usage: Also cited at entry #12.

Original source: "[p. 18, §1.1 "What Is Digital Image Processing?":] "Note that a digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels, and pixels." [p. 66, Ch. 2, Eq. (2-9)–(2-10):] the image is formally represented as an $M \times N$ array $f(x, y)$, and "[e]ach element of this array is called an image element, picture element, pixel, or pel." [p. 702, §10.2 "Point, Line, and Edge Detection":] "Edge pixels are pixels at which the intensity of an image changes abruptly, and edges (or edge segments) are sets of connected edge pixels.""

How to check: Same source as entry #12.

Validated — I have read the original and agree it supports the sentence.

#119 canny1986 — Canny (1986) *NPR Technique Design → V4.2 — Gaussian Prefiltered Sobel*

Sentence in thesis: "Canny showed that the optimal 1D step edge detector under additive noise conditions takes the form of the first derivative of a Gaussian kernel, which is mathematically equivalent to Gaussian smoothing followed by differentiation."

Usage: Also cited at entry #30.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: J. Canny, "A Computational Approach to Edge Detection," *IEEE TPAMI* 8(6), 1986, pp. 679–698, DOI 10.1109/TPAMI.1986.4767851"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#120 roshaan2026 — Roshaan (2026) *NPR Technique Design → V5 — Hierarchical Edge Detection with Multiple Cues*

Sentence in thesis: "Hierarchical edge detection combines multiple complementary boundary cues into a single, unified representation rather than relying on any one signal alone."

Usage: First use of this source.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/NPR/Adaptive Hierarchical Edge Detection.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#121 roberts1963 — Roberts (1963) *NPR Technique Design → V5 — Hierarchical Edge Detection with Multiple Cues*

Sentence in the-sis: "Depth discontinuities recover object silhouettes, surface normal discontinuities reveal geometric structure, and colour gradients computed using the Roberts Cross operator detect texture boundaries that have no corresponding geometric signal."

Usage: Also cited at entry #26.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: L. G. Roberts, *Machine Perception of Three-Dimensional Solids*, PhD thesis, MIT 1963 (Lincoln Lab TR-315), via MIT DSpace"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#122 gonzalez2018 — Gonzalez & Woods (2018) *NPR Technique Design → V5 — Hierarchical Edge Detection with Multiple Cues*

Sentence in the-sis: "Depth discontinuities recover object silhouettes, surface normal discontinuities reveal geometric structure, and colour gradients computed using the Roberts Cross operator detect texture boundaries that have no corresponding geometric signal."

Usage: Also cited at entry #12.

Original source: "[p. 18, §1.1 "What Is Digital Image Processing?":] "Note that a digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels, and pixels." [p. 66, Ch. 2, Eq. (2-9)–(2-10):] the image is formally represented as an $M \times N$ array $f(x, y)$, and "[e]ach element of this array is called an image element, picture element, pixel, or pel." [p. 702, §10.2 "Point, Line, and Edge Detection":] "Edge pixels are pixels at which the intensity of an image changes abruptly, and edges (or edge segments) are sets of connected edge pixels.""

How to check: Same source as entry #12.

Validated — I have read the original and agree it supports the sentence.

#123 kuwahara1976 — Kuwahara et al. (1976) *NPR Technique Design → V6 — Kuwahara Painterly Filter*

Sentence in the-sis: "The Kuwahara filter is a nonlinear smoothing operator that replaces each pixel with the mean colour of whichever overlapping neighbourhood subregion exhibits the lowest internal variance."

Usage: Also cited at entry #31.

Original source: "**Source not available.** Not checked against the original: the PDF is not in the Articles folder. Needed: M. Kuwahara et al., "Processing of RI-Angiocardiographic Images," in *Digital Processing of Biomedical Images*, Plenum Press 1976, pp. 187–203, DOI 10.1007/978-1-4684-0769-3_13"

How to check: NOT CHECKED — primary source missing.

Validated — I have read the original and agree it supports the sentence.

#124 kyprianidis2009 — Kyprianidis et al. (2009) *NPR Technique Design → V6 — Kuwahara Painterly Filter*

Sentence in the-sis: "resolved this by computing a smoothed structure tensor and deriving an elliptical sampling window whose major axis aligns with the dominant direction of local image structure."

Usage: Also cited at entry #32.

Original source: "[Abstract:] "We present a non-photorealistic rendering technique to transform color images and videos into painterly abstractions. It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor." [§3.1 "Orientation and Anisotropy Estimation":] "We calculate the structure tensor directly from the RGB values of the input. . . The structure tensor of f is then defined as" the matrix of pairwise dot products $f_x \cdot f_x$, $f_x \cdot f_y$, $f_y \cdot f_y$, subsequently smoothed with a Gaussian; "Local orientation and a measure for the anisotropy are then derived from the eigenvalues and eigenvectors of the smoothed structure tensor.""

How to check: Same source as entry #32.

Validated — I have read the original and agree it supports the sentence.

#125 demsar2006 — Demšar (2006) *Parameter Adjustment Study*

Sentence in the-sis: "An omnibus Friedman test established whether the ranking within a round differed from chance, and a pairwise Wilcoxon signed-rank test compared the configuration ranked first against its closest competitor."

Usage: First use of this source.

Original source: "[Abstract / Recommendation:] "We recommend Friedman test, with the corresponding post-hoc tests for comparison of more classifiers over multiple data sets." [Sec. 4.1:] "For a two classifier comparison, the Wilcoxon signed ranks test is more appropriate . . . we use average ranks . . . Friedman test . . . followed by post-hoc tests." [Sec. 3.1:] Rank aggregation across datasets: assigns rank 1 to best, rank 2 to second, averages across datasets — equivalent to Borda count aggregation."

How to check: Read directly from `Articles/demsar06a.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#126 mori1970 — Mori (1970) *User Study Design → Objective and Hypotheses*

Sentence in the-sis: "Photorealistic avatars occupy the uncanny valley region where negative affect peaks, while stylised abstraction moves the representation towards the stable ascending slope at lower fidelity."

Usage: Also cited at entry #4.

Original source: "p. 1 ("Valley of familiarity"): "I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. I call this relation the 'uncanny valley.'" Figure 1 (the familiarity-vs-human-likeness graph, x-axis 0–100% human likeness): the curve rises from industrial robot through humanoid robot/stuffed animal, drops sharply through the trough (labelled "corpse" and "prosthetic hand," positioned close to the 100% end of the axis), then rises again to "healthy person" at 100% human likeness, the highest point on the graph (the "second peak"). p. 2 ("The significance of the uncanny"): "In Figure 1, a healthy person is at the top of the second peak.""

How to check: Same source as entry #4.

Validated — I have read the original and agree it supports the sentence.

#127 canales2024 — Canales et al. (2024) *User Study Design → Objective and Hypotheses*

Sentence in the-sis: "Prior work found that stylisation level does not reduce perceived trust when the advisor role and content are held constant."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#128 gao2025 — Gao et al. (2025) *User Study Design → Objective and Hypotheses*

Sentence in the-sis: "Prior work found that stylisation level does not reduce perceived trust when the advisor role and content are held constant."

Usage: Also cited at entry #55.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/IEEE_VR25_TVCG.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#129 alipour2025 — Alipour et al. (2025) *User Study Design → Objective and Hypotheses*

Sentence in the-sis: "Eeriness functions as a specific negative affective state that consistently suppresses positive social evaluations, including trust."

Usage: Also cited at entry #48.

Original source: "The results of our systematic review reveal that most studies rely on static images or hypothetical scenarios with limited real-time interaction, and the majority use subjective trust measures."

How to check: Same source as entry #48.

Validated — I have read the original and agree it supports the sentence.

#130 ho2010 — Ho & MacDorman (2010) *User Study Design → Objective and Hypotheses*

Sentence in the-sis: "Eeriness functions as a specific negative affective state that consistently suppresses positive social evaluations, including trust."

Usage: Also cited at entry #22.

Original source: "[Abstract:] "Phase 2 of this study developed alternative indices with non-significant correlations ($p > .05$) between the proposed y-axis eeriness and x-axis perceived humanness ($r = .02$)." [§3.3.2 Eeriness index:] "Therefore, the final version of the eeriness index would retain eight items that explained 62.04% of the variance and held an overall internal reliability of .74." [§3.3.1 Humanness index:] "Therefore, the final version of the humanness index would retain six items.""

How to check: Same source as entry #22.

Validated — I have read the original and agree it supports the sentence.

#131 mori1970 — Mori (1970) *User Study Design → Objective and Hypotheses*

Sentence in the-sis: "Uncanny valley theory places a real human speaker at the peak of the affinity curve, where human likeness is maximal and comfort is highest."

Usage: Also cited at entry #4.

Original source: "p. 1 ("Valley of familiarity"): "I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. I call this relation the 'uncanny valley.'" Figure 1 (the familiarity-vs-human-likeness graph, x-axis 0–100% human likeness): the curve rises from industrial robot through humanoid robot/stuffed animal, drops sharply through the trough (labelled "corpse" and "prosthetic hand," positioned close to the 100% end of the axis), then rises again to "healthy person" at 100% human likeness, the highest point on the graph (the "second peak"). p. 2 ("The significance of the uncanny"): "In Figure 1, a healthy person is at the top of the second peak.""

How to check: Same source as entry #4.

Validated — I have read the original and agree it supports the sentence.

#132 canales2024 — Canales et al. (2024) *User Study Design → Conditions*

Sentence in thesis: "Canales et al. applied the same principle in their stylisation study, holding speaker identity and scripted content constant across conditions while varying only the avatar rendering style."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#133 bartneck2009 — Bartneck et al. (2009) *User Study Design → Per-Block Questionnaire*

Sentence in thesis: "Part B uses the five-item Godspeed likeability subscale, from Bartneck et al."

Usage: Also cited at entry #21.

Original source: "[Abstract:] "The results have been distilled into five consistent questionnaires using semantic differential scales." [§2:] "In semantic differential scales the respondent is asked to indicate his or her position on a scale between two bipolar words, the anchors. . . a semantic differential format may effectively reduce acquiescence bias without lowering psychometric quality." Figure 1 shows a five point scale ("Strong 1 2 3 4 5 Weak"). Item counts per index are as reported by Ho & MacDorman (2010, §2.1.2), who used the instrument directly: "composed of five indices and 24 semantic differential items. The anthropomorphism index has five items, the animacy index has six items, the likeability index has five items, the perceived intelligence index has five items, and the perceived safety index has three items.""

How to check: Same source as entry #21.

Validated — I have read the original and agree it supports the sentence.

#134 ho2010 — Ho & MacDorman (2010) *User Study Design → Per-Block Questionnaire*

Sentence in the-sis: "Their validation showed that the Godspeed subscales are too strongly intercorrelated to measure distinct constructs, and that no Godspeed subscale independently captures this specific affective state."

Usage: Also cited at entry #22.

Original source: "[Abstract:] "Phase 2 of this study developed alternative indices with non-significant correlations ($p > .05$) between the proposed y-axis eeriness and x-axis perceived humanness ($r = .02$)." [§3.3.2 Eeriness index:] "Therefore, the final version of the eeriness index would retain eight items that explained 62.04% of the variance and held an overall internal reliability of .74." [§3.3.1 Humanness index:] "Therefore, the final version of the humanness index would retain six items.""

How to check: Same source as entry #22.

Validated — I have read the original and agree it supports the sentence.

#135 mayer1995 — Mayer et al. (1995) *User Study Design → Per-Block Questionnaire*

Sentence in the-sis: "**Parts E, F, G — Trust battery (7 point Likert).** Nine items operationalise the Ability-Benevolence-Integrity model of Mayer et al. within the trusting beliefs framework of McKnight and Chervany, introduced in *[cross-ref]*."

Usage: Also cited at entry #19.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/Mayer-IntegrativeModelOrganizational-1995.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#136 mcknight2001 — McKnight & Chervany (2001) *User Study Design → Per-Block Questionnaire*

Sentence in the-sis: "**Parts E, F, G — Trust battery (7 point Likert).** Nine items operationalise the Ability-Benevolence-Integrity model of Mayer et al. within the trusting beliefs framework of McKnight and Chervany, introduced in *[cross-ref]*."

Usage: Also cited at entry #20.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/trust/What Trust Means in E-Commerce Customer Relationships.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#137 mcknight2001 — McKnight & Chervany (2001) *User Study Design → Per-Block Questionnaire*

Sentence in the-sis: "The fourth construct in the McKnight and Chervany typology, predictability, is excluded given that it tends to merge with competence and integrity when participants encounter the advisor for the first time."

Usage: Also cited at entry #20.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/trust/What Trust Means in E-Commerce Customer Relationships.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#138 canales2024 — Canales et al. (2024) *User Study Design → Post-Study Section*

Sentence in the-sis: "P1 is the primary behavioural measure, adapted from the post-study advisor selection paradigm of Canales et al."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#139 alipour2025 — Alipour et al. (2025) *User Study Design → Post-Study Section*

Sentence in thesis: "It captures implicit trust judgment that explicit Likert ratings may not fully reflect, and this addresses the methodological gap identified by Alipour et al."

Usage: Also cited at entry #48.

Original source: "The results of our systematic review reveal that most studies rely on static images or hypothetical scenarios with limited real-time interaction, and the majority use subjective trust measures."

How to check: Same source as entry #48.

Validated — I have read the original and agree it supports the sentence.

#140 mayer1995 — Mayer et al. (1995) *User Study Design → Post-Study Section*

Sentence in thesis: "Mayer et al. identify this disposition as a stable individual difference that must be controlled when comparing trust ratings across conditions."

Usage: Also cited at entry #19.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/Mayer-IntegrativeModelOrganizational-1995.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#141 demsar2006 — Demšar (2006) *User Study Design → Analysis Plan*

Sentence in thesis: "Pairwise Wilcoxon signed-rank tests with Bonferroni correction ($\alpha_{\text{corrected}} = .017$) are then applied to each pair, and C2 versus C3 is the critical pairwise comparison in each case."

Usage: Also cited at entry #123.

Original source: "[Abstract / Recommendation:] "We recommend Friedman test, with the corresponding post-hoc tests for comparison of more classifiers over multiple data sets." [Sec. 4.1:] "For a two classifier comparison, the Wilcoxon signed ranks test is more appropriate . . . we use average ranks . . . Friedman test . . . followed by post-hoc tests." [Sec. 3.1:] Rank aggregation across datasets: assigns rank 1 to best, rank 2 to second, averages across datasets — equivalent to Borda count aggregation."

How to check: Same source as entry #123.

Validated — I have read the original and agree it supports the sentence.

#142 lakens2017 — Lakens (2017) *User Study Design → Analysis Plan*

Sentence in the-sis: "A TOST (two one-sided tests) equivalence test is therefore applied to the C2 versus C3 trust composite, with its bound pre-specified at Cohen's $d = 0.3$, the standard smallest effect size of interest for this procedure."

Usage: New entry, added 2026-08-14. Primary PDF read at `Articles/Lakens -equivalence Test.pdf`. Supports both the TOST procedure and the $d = 0.3$ bound, which the source gives as its own worked example of a SESOI.

Original source: "[Abstract:] "Scientists should be able to provide support for the absence of a meaningful effect. Currently, researchers often incorrectly conclude an effect is absent based a nonsignificant result. A widely recommended approach within a frequentist framework is to test for equivalence. In equivalence tests, such as the two one-sided tests (TOST) procedure discussed in this article, an upper and lower equivalence bound is specified based on the smallest effect size of interest." [p. 356:] "A very simple equivalence testing approach is the 'two one-sided tests' (TOST) procedure (Schuirmann, 1987). In the TOST procedure, an upper (Δ_U) and lower (Δ_L) equivalence bound is specified based on the smallest effect size of interest (SESOI; e.g., a positive or negative difference of $d = .3$)."""

How to check: Open `Articles/Lakens -equivalence Test.pdf`, abstract and p. 356, left column.

Validated — I have read the original and agree it supports the sentence.

Chapter: Implementation

#143 roshaan2026 — Roshaan (2026) *NPR Technique Implementation → Meta Avatars SDK*

Sentence in the-sis: "The normal layer reuses the world-space normal derivative computation from V3 (*[cross-ref]*), and an AHEAD-style adaptive gain then amplifies this layer specifically in darker image regions."

Usage: Also cited at entry #118.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/NPR/Adaptive Hierarchical Edge Detection.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#144 kyprianidis2009 — Kyprianidis et al. (2009) *NPR Technique Implementation → V6 — Kuwahara Painterly Filter*

Sentence in the-sis: "All three platforms use the anisotropic Kuwahara filter of Kyprianidis et al."

Usage: Also cited at entry #32.

Original source: "[Abstract:] "We present a non-photorealistic rendering technique to transform color images and videos into painterly abstractions. It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor." [§3.1 "Orientation and Anisotropy Estimation":] "We calculate the structure tensor directly from the RGB values of the input… The structure tensor of f is then defined as" the matrix of pairwise dot products $f_x \cdot f_x$, $f_x \cdot f_y$, $f_y \cdot f_y$, subsequently smoothed with a Gaussian; "Local orientation and a measure for the anisotropy are then derived from the eigenvalues and eigenvectors of the smoothed structure tensor.""

How to check: Same source as entry #32.

Validated — I have read the original and agree it supports the sentence.

#145 kyprianidis2009 — Kyprianidis et al. (2009) *NPR Technique Implementation → Meta Avatars SDK*

Sentence in the-sis: "The exact weighting formula differs by platform, since each implementation follows a different variant of the same principle from Kyprianidis et al."

Usage: Also cited at entry #32.

Original source: "[Abstract:] "We present a non-photorealistic rendering technique to transform color images and videos into painterly abstractions. It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor." [§3.1 "Orientation and Anisotropy Estimation":] "We calculate the structure tensor directly from the RGB values of the input… The structure tensor of f is then defined as" the matrix of pairwise dot products $f_x \cdot f_x$, $f_x \cdot f_y$, $f_y \cdot f_y$, subsequently smoothed with a Gaussian; "Local orientation and a measure for the anisotropy are then derived from the eigenvalues and eigenvectors of the smoothed structure tensor.""

How to check: Same source as entry #32.

Validated — I have read the original and agree it supports the sentence.

Chapter: Evaluation

#146 mori1970 — Mori (1970) *User Study Results → H1 — NPR Reduces Eeriness*

Sentence in the-sis: "H1 was motivated by the uncanny valley account: a photorealistic avatar occupies the near-human fidelity region of the affinity curve, where negative affect peaks."

Usage: Also cited at entry #4.

Original source: "p. 1 ("Valley of familiarity"): "I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. I call this relation the 'uncanny valley."' Figure 1 (the familiarity-vs-human-likeness graph, x-axis 0–100% human likeness): the curve rises from industrial robot through humanoid robot/stuffed animal, drops sharply through the trough (labelled "corpse" and "prosthetic hand," positioned close to the 100% end of the axis), then rises again to "healthy person" at 100% human likeness, the highest point on the graph (the "second peak"). p. 2 ("The significance of the uncanny"): "In Figure 1, a healthy person is at the top of the second peak.""

How to check: Same source as entry #4.

Validated — I have read the original and agree it supports the sentence.

#147 canales2024 — Canales et al. (2024) *User Study Results → H2 — NPR Does Not Reduce Trust*

Sentence in the-sis: "The prediction was grounded in results from Canales et al., who found that trust ratings were broadly equivalent across three levels of stylisation in a scripted advisor paradigm."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#148 gao2025 — Gao et al. (2025) *User Study Results → H2 — NPR Does Not Reduce Trust*

Sentence in thesis: "that realistic and stylised agents attract comparable levels of confidence."

Usage: Also cited at entry #55.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/IEEE_VR25_TVCG.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.

#149 alipour2025 — Alipour et al. (2025) *User Study Results → H3 — Eeriness Predicts Trust*

Sentence in thesis: "This result matches the systematic review of Alipour et al., who found a direct negative relationship between eeriness and trust in the majority of studies that examined the association."

Usage: Also cited at entry #48.

Original source: "The results of our systematic review reveal that most studies rely on static images or hypothetical scenarios with limited real-time interaction, and the majority use subjective trust measures."

How to check: Same source as entry #48.

Validated — I have read the original and agree it supports the sentence.

#150 canales2024 — Canales et al. (2024) *User Study Results → Post-Study Rankings*

Sentence in thesis: "A rank order was used rather than the strict forced-choice employed by Canales et al."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#151 canales2024 — Canales et al. (2024) *User Study Results → Post-Study Rankings*

Sentence in the-sis: "observed a dissociation between explicit and behavioural measures in their related study: scores on the trust scales were equivalent across stylisation levels, yet the avatar at intermediate realism attracted the highest selection rate."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#152 canales2024 — Canales et al. (2024) *User Study Results → Post-Study Rankings*

Sentence in the-sis: "found for their overall preference measure."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#153 canales2024 — Canales et al. (2024) *User Study Results → Secondary Measures*

Sentence in the-sis: "similarly found no significant relationship between propensity and trust ratings in their comparable manipulation."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

Chapter: Conclusion

#154 mori1970 — Mori (1970) *Summary of Contributions*

Sentence in the-sis: "H3 was confirmed: eeriness negatively predicted every trust subscale within participants, replicating the mechanism the uncanny valley account proposes."

Usage: Also cited at entry #4.

Original source: "p. 1 ("Valley of familiarity"): "I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. I call this relation the 'uncanny valley."' Figure 1 (the familiarity-vs-human-likeness graph, x-axis 0–100% human likeness): the curve rises from industrial robot through humanoid robot/stuffed animal, drops sharply through the trough (labelled "corpse" and "prosthetic hand," positioned close to the 100% end of the axis), then rises again to "healthy person" at 100% human likeness, the highest point on the graph (the "second peak"). p. 2 ("The significance of the uncanny"): "In Figure 1, a healthy person is at the top of the second peak.""

How to check: Same source as entry #4.

Validated — I have read the original and agree it supports the sentence.

#155 kyprianidis2009 — Kyprianidis et al. (2009) *Limitations*

Sentence in the-sis: "**Unsmoothed structure tensor estimation.** The anisotropic Kuwahara formulation of Kyprianidis et al."

Usage: Also cited at entry #32.

Original source: "[Abstract:] "We present a non-photorealistic rendering technique to transform color images and videos into painterly abstractions. It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor." [§3.1 "Orientation and Anisotropy Estimation":] "We calculate the structure tensor directly from the RGB values of the input. . . The structure tensor of f is then defined as" the matrix of pairwise dot products $f_x \cdot f_x$, $f_x \cdot f_y$, $f_y \cdot f_y$, subsequently smoothed with a Gaussian; "Local orientation and a measure for the anisotropy are then derived from the eigenvalues and eigenvectors of the smoothed structure tensor.""

How to check: Same source as entry #32.

Validated — I have read the original and agree it supports the sentence.

#156 canales2024 — Canales et al. (2024) *Limitations*

Sentence in the-sis: "**Trust subscale item count.** The trust subscales use three items per dimension, adapted from the single-item measures of Canales et al."

Usage: Also cited at entry #6.

Original source: "Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""

How to check: Same source as entry #6.

Validated — I have read the original and agree it supports the sentence.

#157 mcknight2001 — McKnight & Chervany (2001) *Limitations*

Sentence in the-sis: "**Trust subscale item count.** The trust subscales use three items per dimension, adapted from the single-item measures of Canales et al. and grounded in the McKnight & Chervany."

Usage: Also cited at entry #20.

Original source: "**Passage not copied yet.** The source is available but the exact supporting wording has not been quoted here."

How to check: Open `Articles/trust/What Trust Means in E-Commerce Customer Relationships.pdf` and locate the passage supporting the sentence above.

Validated — I have read the original and agree it supports the sentence.
